

Q-TUNE-F10-SH

E-O Q-switched diode-pumped tuneable wavelength laser

TECHNICAL DESCRIPTION

&

USER'S MANUAL

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1 Warranty Statement

Quantum Light Instruments Ltd. (legal name UAB “Quantum Light Instruments”), further – QLI - warrants to the original purchaser that laser devices are free from defects in parts and workmanship. QLI will make any necessary repairs or replacement of parts to remedy any defect according to the conditions drawn up in the contract.

The foregoing warranty does not cover equipment that is damaged by accident or improper use. QLI does not assume any liability if adaptations are made or accessories attached to the equipment that impair or alter the normal functioning of the equipment. The limited warranty and remedy contained in this paragraph are the only warranty and remedy pertaining to the equipment. QLI DISCLAIMS ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. *QLI* is *not* liable for any accidental, consequential or other damages or costs, lost profits or inconvenience occasioned by loss of the use of the equipment or labor expended by persons not so authorized by *QLI*.



2 *Laser Safety*

2.1 *Safety Features and Compliance to Government Requirements*

This laser is a Class IV (Class 4) laser product according to radiation danger degree, and, by definition, relates to certain safety and fire hazards. The following features are incorporated into the laser to conform to several government requirements. The applicable United States Government requirements are contained in 21 CFR, chapter 1, subchapter J, administered by the Center for Devices and Radiological Health (CDRH). The European Community requirements for product safety are specified in the Low Voltage Directive (LVD) **2014/35/EU**. The Low Voltage Directive requires that lasers comply with the **EN60825-1:2014** (Radiation Safety of Laser Products) and **EN61010-1:2010** (Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements).

The laser head is enclosed in a protective housing that prevents human access to radiation in excess of the limits of Class 1 radiation as specified in 21 CFR, subchapter J, Section 1040.10(f) (1) and Tables 3 and 4 of EN60825-1, except for the output beam, which is Class IV (Class 4) (as specified in Table 8 of EN60825-1, “Accessible emission limits for Class 3B laser products”).

The appropriately labeled light on the laser head illuminate before laser emission can occur. Amber light is used so that it will be seen when the proper type of safety glasses are used (21 CFR, subchapter J, Section 1040.10(f) (5) /EN60825-1, clause 6.7).

The laser controls are positioned so that the operator is not exposed to laser emission while manipulating the controls (21 CFR, subchapter J, Section 1040.10(f) (7) /EN60825-1, clause 6.9).

2.2 *Electromagnetic Compatibility*

Q2 series laser system complies with the European requirements for electromagnetic compatibility as defined in the Electromagnetic Compatibility Directive **2014/30/EU**.

This laser system is intended for use in an ISM (Industrial, Scientific and Medical) Environment. Operation of this laser system in a different EMC environment may require that the user take remedial action in addition to the normal installation and operation described in this manual to resolve potential electromagnetic compatibility problems. *QLI* makes no claims beyond those listed below concerning the compatibility of this laser system in EMC environment other than ISM environment.



2.3 Labeling

Table 1: Laser safety labels

	IEC Hazard symbol
	IEC Aperture label
	CE warning label
	IEC Class 4 warning label
	High voltage hazard label
	IEC hazard to skin label



 Quantum Light Instruments Ltd. Mokslininku 6A, Vilnius, 08412, Lithuania MANUFACTURED: MONTH _____ YEAR _____ MODEL _____ SERIAL# _____ THIS LASER PRODUCT COMPLIES WITH 21 CFR 1040 AS APPLICABLE EXCEPT FOR DEVIATIONS PURSUANT TO LASER NOTICE NO.50, 07.26.2001	Identification/certification label
	CE power supply hazard label
	Earth (ground) TERMINAL label

2.4 Location of Safety Labels and Laser Emission Indicator

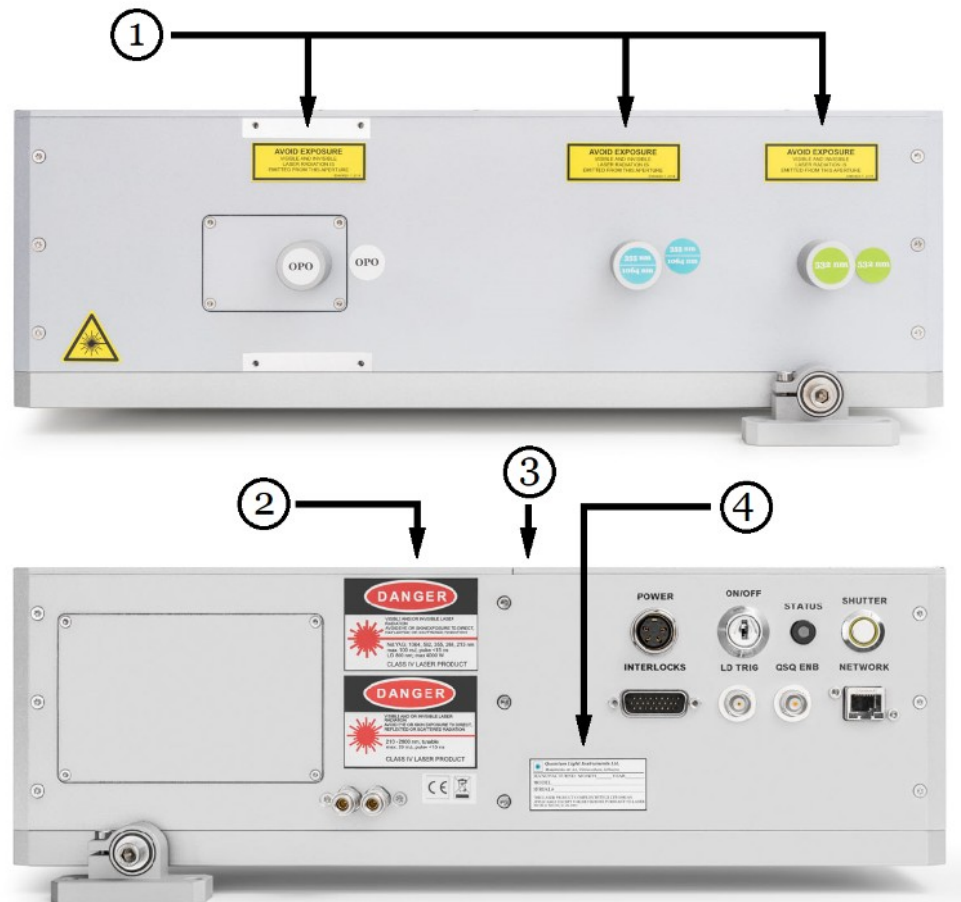


Figure 1: Q2 laser head safety labeling: 1) Aperture labels, 2) Warning labels, 3) Laser emission indicator, 4) Identification/Certification label



2.5 *Precautions for the Safe Operation of Class IV Lasers*

- Wear protective eyewear at all times; selection depends on the wave- length and intensity of the radiation, the conditions of use, and the visual function required.
- Maintain a high ambient light level in the laser operation area. This keeps the eye's pupil constricted, thus reducing the possibility of eye damage.
- Keep the protective cover on the laser at all times.
- Avoid looking at the output beam; even diffuse reflections are hazardous.
- Avoid wearing jewelry or other objects that may reflect or scatter the beam while using the laser.
- Use an infrared detector or energy detector to verify that the laser beam is off before working in front of the laser.
- Operate the laser at the lowest beam intensity possible, given the requirements of the application.
- Expand the beam whenever possible to reduce beam power density.
- Avoid blocking the output beam or its reflection with any part of your body.
- Care should be taken to avoid possible external back reflections to the laser cavity. Back reflections can damage laser components which can result in the immediate decrease of the laser power or reduction of their longevity. To avoid this use energy absorbing target (such as beam trap) to capture the laser beam and prevent unnecessary reflections or scattering. It is recommended to use beam trap shown in Figure 2 or its equivalent.



Figure 2: Beam trap Thorlabs, model BT610/M



3 Environmental Specifications

3.1 CE Electrical Equipment Requirements

For information regarding the equipment needed to provide the electrical service listed under “Operating Requirements” at the end of [chapter 4](#), please refer to EU standard EN60309-1, “Plugs, socket-outlets and couplers for industrial purposes”.

3.2 Environmental Specifications

Environmental conditions under which the laser system will function are listed below:

Indoor use only.

Altitude: up to 2000 m.

Temperatures: 15° C to 30° C.

Maximum relative humidity: 80 % non-condensing for temperatures up to 30° C.

Mains supply voltage: do not exceed $\pm 10\%$ of the nominal voltage stated in [chapter 4](#).



4 Specifications ¹⁾

Table 2: Laser specifications

Model	Q-TUNE				
	-C10	-C100	-E10	-E33	-F10
Wavelength range, nm ²⁾ OPO SH extension	410 - 2300 nm 210-410 nm				
Pulse repetition rate ³⁾	10 Hz	100 Hz	10 Hz	33 Hz	10 Hz
Pulse energy ⁴⁾	>1 mJ		>5 mJ		>8 mJ
Linewidth	<6 cm ⁻¹				
Pulse duration ⁵⁾	< 5 ns				
Pulse-to-pulse stability ⁶⁾	< 4.5 % RMS				
Power drift ⁷⁾	± 3.0 %				
Polarization	vertical				
Typical beam diameter ⁸⁾	3 mm		5 mm		6 mm
Typical beam divergence ⁹⁾	< 3 mrad		< 5 mrad		
Jitter ¹⁰⁾	< 0.5 ns RMS				
Pump laser outputs ¹¹⁾					
Max pulse energy at ¹²⁾					
Fundamental	15 mJ		50 mJ		65 mJ
2 nd harmonic	7 mJ		20 mJ		30 mJ
3 rd harmonic	5 mJ		20 mJ		30 mJ
Dimensions					
Laser head (W×L×H)	390× 620 × 135 mm ³		390× 620 × 153 mm ³		
Power adapter (W×L×H) ¹³⁾	50×125×33 mm ³		192 × 178 × 46 mm ³		
APU unit (with +28VDC source integrated) (W×L×H) ¹⁴⁾	89 × 422 × 330 mm ³				
Operating requirements					
Cooling requirements	Air cooled (water-free)				
Ambient temperature	15 – 25 °C				
Relative humidity (non-condensing)	10 – 80 %				
Mains voltage	90-230 VAC, single phase, 47-63 Hz ¹³⁾				
Average power consumption	< 50 W	<150 W	<100 W	<150 W	

1) Due to continuous improvements all specifications are subject to change. Unless stated otherwise all specifications are measured at 450 nm and max pulse repetition rate. The parameters marked typical are not specifications. They are indications of typical performance and will vary with each unit we manufacture

2) Extension down to 210 nm is available with optional second harmonic generator.

3) Factory-set pulse repetition rate in internal triggering mode. Pulse repetition rate can be divided-down to 1 Hz.

4) Measured at 450 nm output. See tuning curves for pulse energies at other wavelengths.

5) FWHM level at 1053/1064 nm, measured with 350 ps rise time photodiode.

6) Measured during 30 seconds operation after warm-up.

7) Over 8 hour period after 20 minutes of warm-up, when ambient temperature variation is less than ±2 °C. Power value is calculated every 1 second.

8) Beam diameter is measured 20 cm from laser output at the 4σ level.

9) Full angle measured at the 4σ level.

10) In respect to falling edge of pump diode triggering pulse.

11) Laser pulse energy is optimized for OPO pumping and may differ from stand-alone laser specifications.

12) Outputs can be configured for simultaneous or non-simultaneous with OPO operation. Values indicated here are for non-simultaneous operation and are typical.

13) Laser can be powered from appropriate 12 or 27 VDC power source, depending on model. Please inquire for details.



5 Dimensions

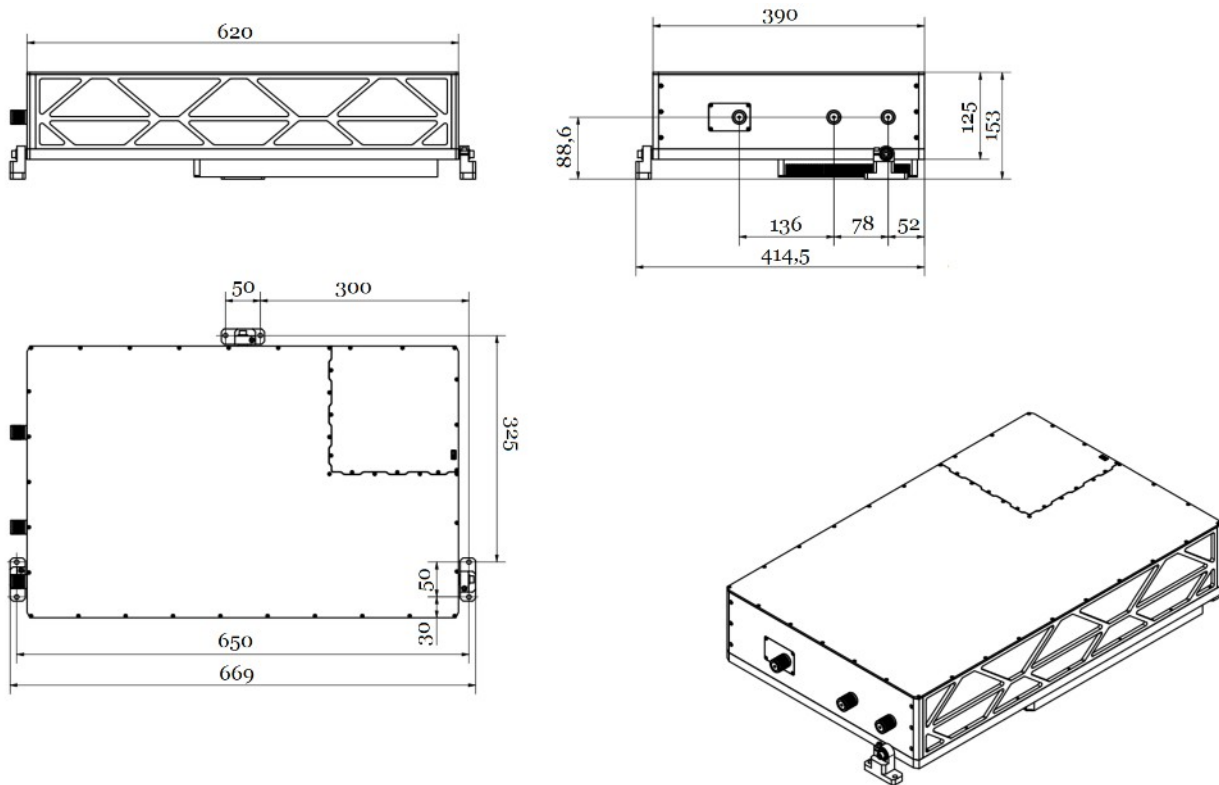


Figure 3: Q-Tune laser head dimensions (mm)

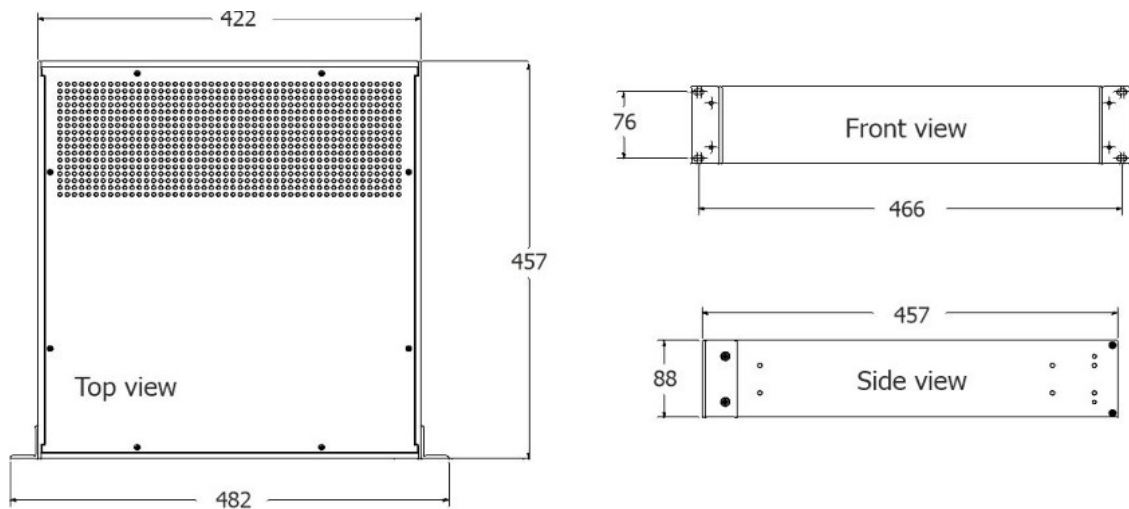


Figure 4: Q-Tune laser APU unit with integrated power supply, standard 19'' rack 2U height housing.



6 Delivery set

Q-TUNE-laser head	1
Air purging unit (APU)	1
Power supply	1
Cable and hose set	1
Crossover LAN cable	1
USB flash drive with user manual	1
USB – Ethernet adapter	1



7 Laser Description

Tunable wavelength laser Q-TUNE integrates pump laser, harmonics generators H-SMART and Optical Parametric Oscillator (OPO).

The general optical layout of the laser is presented in Figure 5

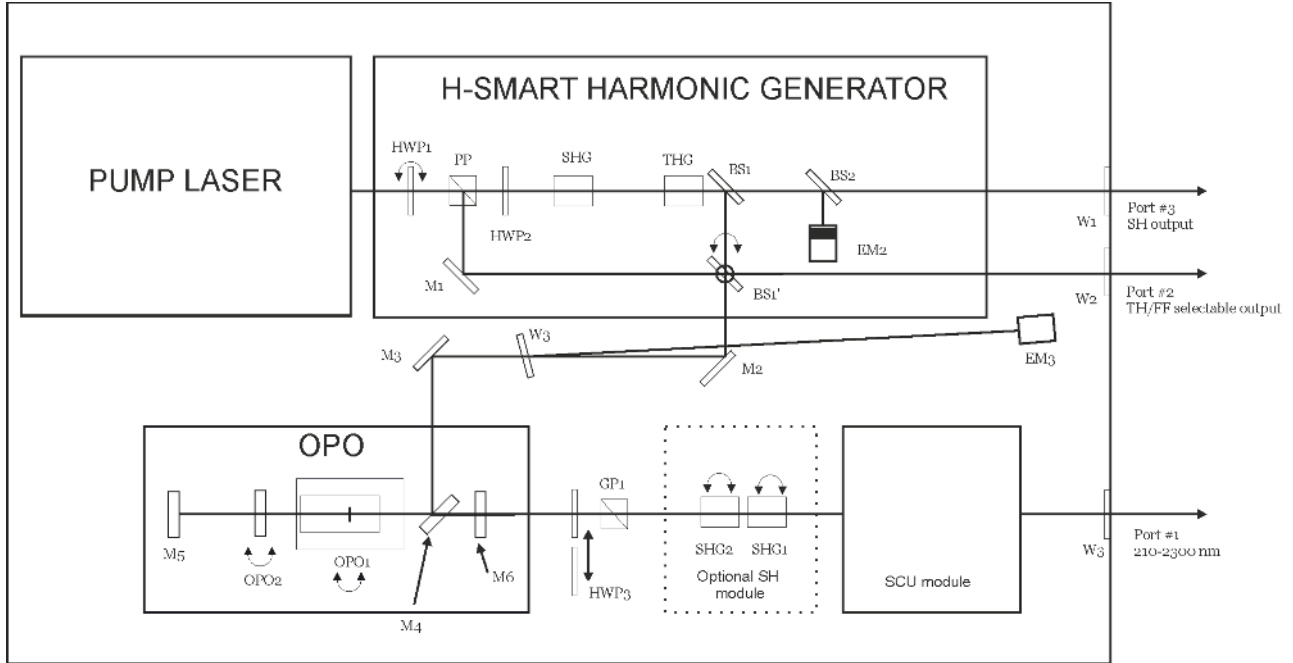


Figure 5: Optical layout of Q-TUNE laser.

Heart of any laser system is master oscillator. The Q2 laser is compact diode-pumped solid state actively Q-switched laser that produces infrared beam with duration of less than 5 ns, with the pulse repetition rate in 10-100 Hz range.

Optical layout of laser is presented in Figure 6.

The master oscillator is designed to produce large mode volume for maximum energy extraction from laser rod. Laser rod LR1 is pumped by two high power laser diode bars LD1 and LD2. The pump bar radiation is shaped to produce nearly round pump beam profile inside laser crystal for optimal resonator mode overlap. Laser cavity consists from rear mirror RM and output coupler OC. Folding mirrors M1 and M2 are used to guide pump radiation to the active element. Active Q-switching is realized by an electro-optical modulator based on the Pockels effect PC and polarizer P1.



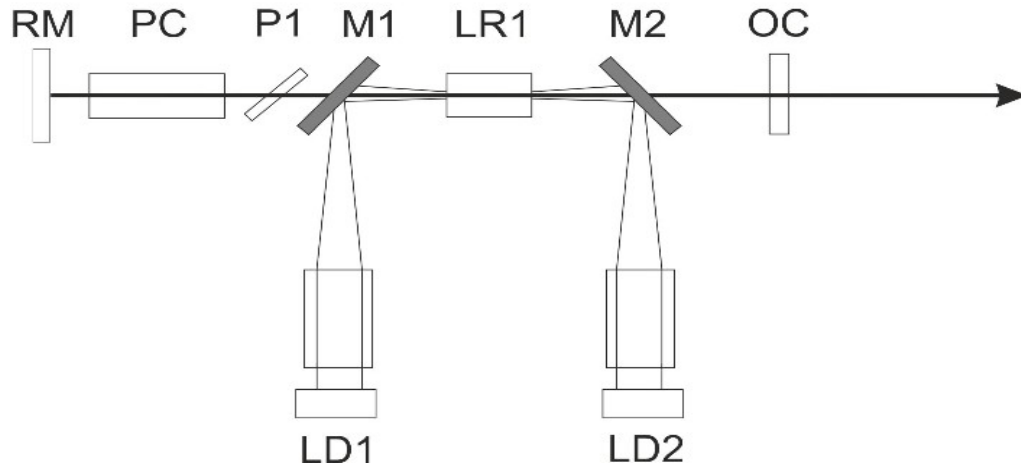


Figure 6: Optical layout of Q2 laser. RM- rear mirror, PC – Pockels cell, P1 - polariser, M1, M2 – mirrors, LR1 – laser rod, OC – output coupler, LD1, LD2 - pump diodes.

Master oscillator produces fundamental media radiation wavelength at 1064 nm for Nd:YAG and 1053 nm for Nd:YLF laser crystals. In the next step, the fundamental radiation is converted into the second (532 or 527 nm) and third (355 or 351 nm) harmonics in H-SMART harmonics generator.

Harmonic wavelengths are generated by nonlinear crystal SHG and THG and then required wavelength is separated from residual output by beam separators BS1, BS1' and directed to the Q-TUNE output or for OPO pumping.

PC controlled switching between fundamend/second/third harmonic output is arranged with the help of motorized HWP1 waveplate, PP polariser and movable mirror BS3. When fundamental radiation polarization is vertical, the beam is reflected by PP1 polariser to M1 mirror and to the output of unit.

By rotating HWP1 waveplate polarization of fundamental beam can be made horizontal. In this case beam is transmitted by PP polariser to the SHG and THG crystals and third harmonic output is produced. Movable mirror BS1' can direct third harmonic beam to the Q-TUNE output or allow it to pump OPO unit.

Second harmonic beam is filtered from fundamental by BS1 beam splitter and directed to module output.

Energy monitors of second and third harmonics and OPO beams are used for self-tuning of harmonic crystals temperatures for maximum OPO output.

The optical parametric oscillator is a solid state continuously tunable source of visible and near IR radiation. Based on BBO nonlinear crystals, the OPO produces tuneable wavelength beam in the range of 410-2300 nm.



The pumping beam is directed by dichroic mirror M2 and M3 to the OPO cavity. OPO cavity is formed by mirrors M5 and M7. Wavelength tuning is achieved by rotation of nonlinear crystals OPO1.

OPO output wavelength can be doubled by **optional** second harmonic (SHG) crystals, producing output in 210-409 nm range.

Signal and idler waves are separated by the Glan prism GP1 and dispersive components based SCU unit.

Electrical interconnection of Q-TUNE is shown in Figure 7.

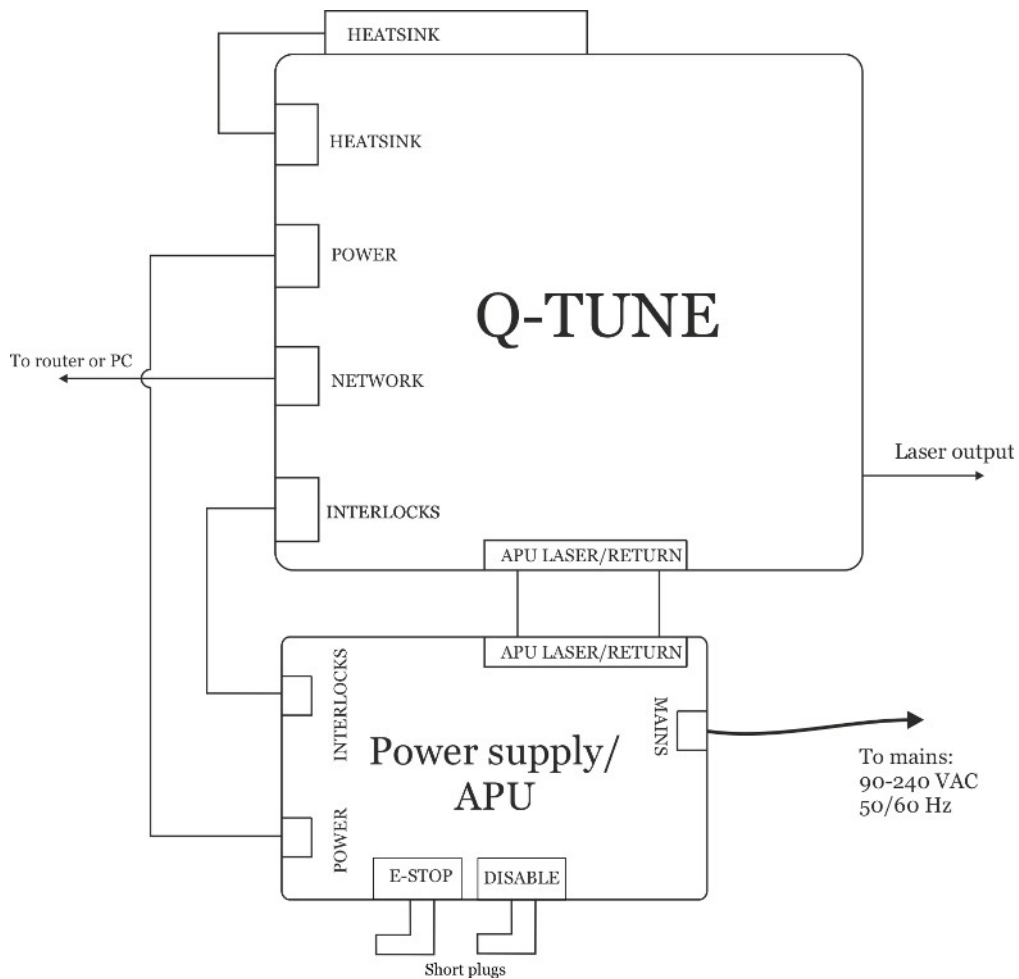


Figure 7: Electrical connections of Q-TUNE laser



8 Installation

Laser and auxiliary units must be settled in a place void of dust and aerosols, where exploitation conform to the regulations specified in [chapter 3.2](#) of present manual and comply with below conditions:

Lasers work best in a temperature stabilized environment. Therefore, it is advisable to operate the laser in an air-conditioned room, provided that the laser is placed at a distance from air conditioning outlets.

The laser should be positioned on an optical table or breadboard. The laser mounting is most conveniently done on optical tables with mounting holes on 25 mm grid.

8.1 Laser Installation Procedure

All personnel in the area must wear laser goggles/glasses of the approved type at all times during operation of the laser. This protective eyewear must be effective at the wavelengths generated from the laser. It is assumed that the operator has read Safety chapter and is familiar with laser safety practices and the dangers involved.

This chapter describes installation procedure of standard Q-TUNE series laser that is delivered with controller unit.

1. Inspect laser boxing for damage caused by transportation. If any damage is present, inform QLI immediately.
2. To avoid condensation forming after bringing the laser from the cold storage or transportation site, allow the box to warm up to room temperature (approx. 3-4 hours) before opening it.
3. Unpack the laser. Inspect it for presence of all components and for damage caused by transportation. When any of such are present, inform QLI immediately.
4. **Remove laser output or/and input window seals used for protection against dust during transportation.**
5. Fix the laser head to the optical table. Place Air Purging Unit (APU) near the laser head in a convenient place.
6. Connect cable to the “POWER”(marking may differ depending on the model) socket on the rear laser panel (see Figure 8) and circular connector marked “LASER HEAD (POWER)” on the rear panel of APU unit (Figure 9).
7. Connect DB connectors marked “INTERLOCKS” on the laser head with “INTERLOCKS/HU/ATTN” on the rear panel of APU unit (Figure 8 and Figure 9).
8. Connect laser heatsink located underneath the laser body with “HEATSINK” connector on the rear panel of the laser.
9. Connect blue air hoses to laser rear panel and APU unit as is shown in the Figure 9 and Figure 8. “LASER” on APU unit should be connected to “LASER” on laser head etc.

10. Connect the mains cable to the APU unit. Connect cable to the AC mains (100-230 VAC). Switch POWER ON/OFF button, located on the front panel of APU unit into ON position (Figure 10). The STATUS LED on the rear panel of Q-TUNE should turn green, indicating that power is present. Some models of mains adapters have ON/OFF switch that is required to be switched into ON position.

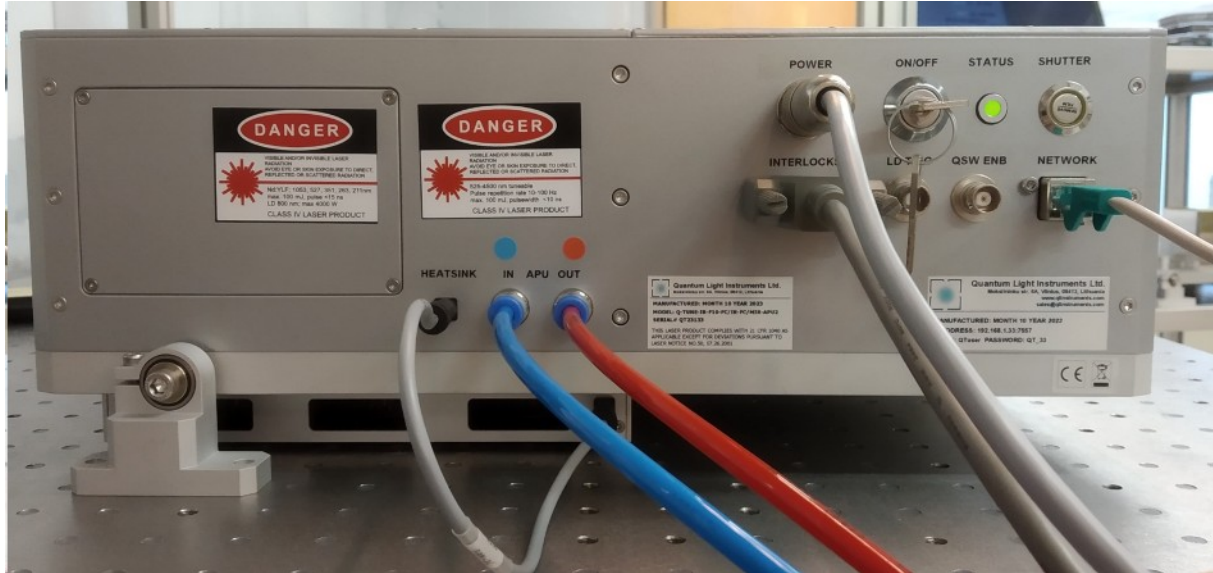


Figure 8: Q-TUNE laser with rear panel cables connected.



Figure 9: Rear panel of APU unit with cables interconnected.

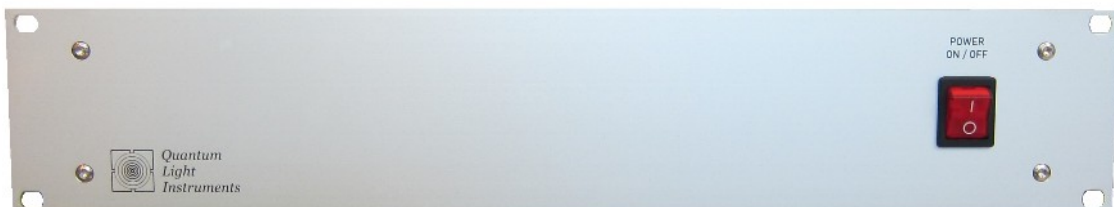


Figure 10: Front panel of APU unit



11. Nonlinear crystals in harmonics generator and OPO cavity needs 10-20 min to reach operating temperatures. Wait minutes until crystal temperatures are stabilized.

Note:

Do not disconnect the APU unit from the mains and do not switch the POWER OFF. The harmonics crystals should be constantly heated for long lifetime.

12. Turn the key switch on the rear panel of Q-TUNE clockwise (position “ON”). The STATUS LED should turn red, then after ~15 seconds should start blinking green or turn on orange. Blinking green/yellow/red indicates that laser cavity is warming up. Once laser cavity temperature reaches required value (typically in 26-34 deg Celsius range), the STATUS LED changes color to yellow.
13. Arrange laser connection to Personal Computer(PC). Please check [Chapter 11](#) for Ethernet port configuration details.
14. Connect to Q-TUNE laser control web-page. From “Outputs” tab activate required port output and/or set required wavelength of the laser. Switching from one wavelength to another requires less than 1 sec typically, however, maximizing second harmonic output might require ~5-10 minutes due to crystal temperature change and stabilization.

Note:

During harmonic generator reconfiguration, laser operation might be temporarily disabled until crystal temperatures reach preset values.

15. Check pulse energy of fundamental laser output. For this, activate FF output from “Outputs” tab by pushing button “Max FF output”.
16. Place energy meter at output port #2 marked “355/1064 nm”. Start the laser and measure pulse energy.
17. If pulse energy is lower than specified value, contact QLI representative for further instructions. Check “Test_Data” file for factory measured values, file can be downloaded from HELP tab.
18. Next, check third harmonic output at the same port (#2) marked “355/1064 nm”. For this, push “Max TH output”.
19. If pulse energy is lower than specified value (check “Test_Data” file for factory measured values), contact QLI representative for further instructions.
20. Next, check second harmonic output at port #3 marked “532 nm”. For this, push “Max SH output”.
21. If pulse energy is lower than specified value (check “Test_Data” file for factory measured values), contact QLI representative for further instructions.
22. Next, check OPO output pulse energy. For this, place power meter at the exit of Port #1, marked “OPO”. Push “Enable OPO” button on “Outputs” tab. Enter 450 nm wavelength in the “Set Wavelength” field and push “Set” button.
23. If pulse energy is lower than specified value (check “Test_Data” file for factory measured values), contact QLI representative for further instructions.
24. Stop the laser by releasing Shutter ON/OFF button. Initial laser installation is completed.



9 Laser Control Interfaces

This chapter describes control interfaces of the laser that are available for OEMs. Laser control sockets are located on laser rear panel and laser controller (see Figure 11). 4-pin “Power” connector provides power to the laser from DC power supply or AC/DC adapter. "INTERLOCKS" D-SUB 15 pin socket allows laser control described in table below.

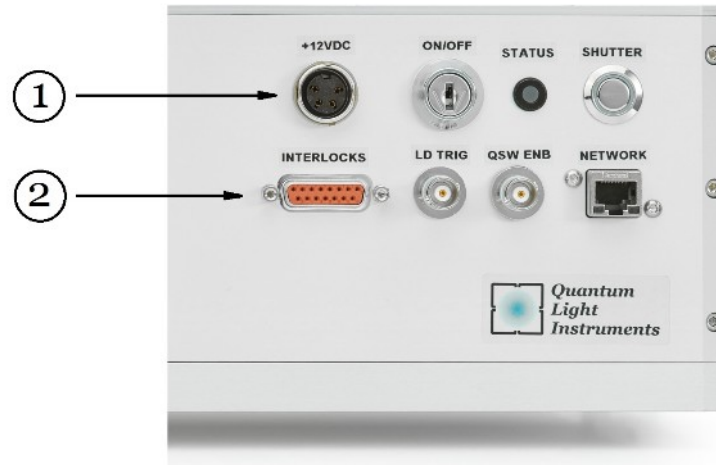


Figure 11: Laser control socket location: 1) Power 2) Interlock

Table 3: Assignment of “Power” socket pins

Pin No.	Description
1, 2	Power supply ground
3, 4	+28 VDC from power supply



Table 4: Assignment of "INTERLOCKS" socket pins

Pin #	Label	Type	Description
1, 9, connector shield	INTERLOCK RET	Analog GND	Return GND for DISABLE and/or ESTOP inputs.
2	DISABLE	INPUT 2kOhm pull-up to internal 5V	Non-latching safety interlock input. Apply 3.7V .. 5V, or leave unconnected to disable laser emission. Drive this pin low or make short to "INTERLOCK RET" pin in order to enable laser emission. This input can be used to interlock laser for repetitive laser start – stop operation.
3, 4	RELAY	Floating relay contacts	Dry contacts of internal relay that are closed when keyswitch is in "ON" position. Relay is rated for 1 A, 28 VDC. These pins can be used for warning sign activation as an example.
5, 6, 14		Reserved	Reserved for internal use. Leave these pins intact or not connected.
7		Reserved	Reserved for models with "APU" unit: power output for APU pump.
8, 11	12V RET	Power GND	Return GND for 12V and "LASER READY" outputs.
10	E-STOP	INPUT 2kOhm pull-up to internal 5V	Latching safety interlock input. Apply 3.7V .. 5V, or leave unconnected to disable laser emission permanently, until mains power supply is cycled. Drive this pin low or make short to "INTERLOCK RET" pin prior to mains power-up in order to enable laser emission.
12	12V CONT	Power OUTPUT 300mA MAX	Continuous 12V power for user.
13	LASER READY	OUTPUT Open drain, 12V/1A MAX	Open drain output which is activated when laser is ready for operation. External pull-up resistor is required to use this output. Use "12V RET" power GND pin as a return path for this open drain output.
15		Reserved	Reserved for models with "APU" unit: return path for APU pump.



10 Routine operation

This section provides basic operation instructions for the Q-TUNE laser including powering up, operating, pausing, and shutting down.

All personnel in the area must wear laser goggles/glasses of the approved type at all times during operation of the laser. This protective eyewear must be effective at the wavelengths generated by the laser. It is assumed that the operator has read Safety chapter and is familiar with laser safety practices and the dangers involved.

The front panel laser controller is shown in Figure 12. The purpose of controls is described in Table 5.



Figure 12: Rear panel controls of Q-TUNE



Table 5: Descriptions of front panel controls

Control	Description
ON/OFF Keyswitch	Enables laser operation. Removal of the key prevents unauthorized laser operation.
SHUTTER button	Starts or stops lasing at preset pulse repetition rate. Default pulse repetition rate is 10 Hz. Laser emission is indicated by yellow illumination of the button.
STATUS	Multicolor LED has following states: 1. Green – Stand-by mode (mains are connected, Keyswitch in off position). 2. Green-yellow-red blinking- device is warming up. The laser body temperature is not in the operational range yet. 2. Orange continuous – Ready mode (key-switch is in ON position, laser is not firing). 3. Orange fast blinking – Laser is operating. 4. Orange slow blinking – Laser is in external triggering mode and waiting for external triggering pulse to be applied. 5. Red/red fast blinking – device is in configuration mode 6. Red slow blinking - Error of operation. Please check error code from web-interface and contact manufacturer for information.
LD TRIG	Bidirectional input/output. Socket is of BNC type. In internal triggering mode provides positive TTL level pulse that has rising edge coincident with start of pump diode and falling edge coincident with Q-switch activation. In external triggering mode the input accepts TTL triggering pulse for starting of pump diodes. The falling edge of pulse activates Q-switch. Input impedance is approx. 50 Ohm.
QSW ENB	Q-SW trigger output. Socket is of BNC type. Internal triggering mode provides positive TTL level pulse that has rising edge approximately coincident with laser pulse. When in external triggering mode, high level on this input disables Q-Switch operation.
NETWORK	RJ-45 type socket for connection of controller to router of PC.



10.1 Laser Triggering Modes

NOTE: Nd:YLF and Nd:YAG lasers has different triggering pulse requirements in single pulse mode.

Laser can be operated in internal or external triggering mode. Triggering mode configuration can be changed from web-interface only. See [chapter 11](#) for instructions how to control laser from its web-interface. Triggering mode control is available from “Main” window.

10.1.1 Internal Triggering

In internal triggering mode required timing pulses are produced by internal pulse generator. Internal timing pulses are accessible from LD TRIG and QSW ENB sockets on the front panel of laser controller. Timing chart is shown in Figure 13.

Q1/Q2/Q2HE/Q-SPARK/Q-TUNE laser timing diagrams (internal)

Note: Time axis scale is variable for clearer presentation.

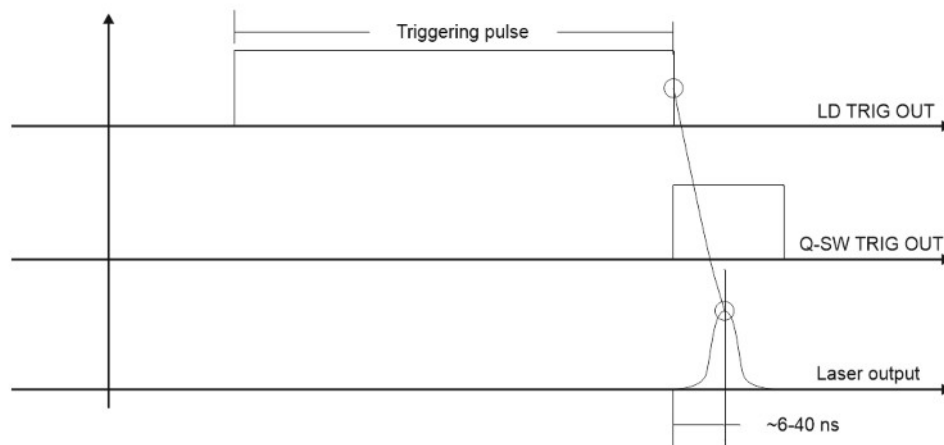


Figure 13: Timing chart of Q2 laser in internal triggering mode

In the internal triggering mode at LD TRIG socket positive TTL level pulse is produced with its rising edge coincident of start of pump diode activation, and falling edge coincident with Q-Switch activation. Laser pulse is produced with a delay of 6-40 ns in respect to falling edge of LD TRIG pulse. Laser pulse jitter is less than 0.5 ns RMS.

QSW ENB output produces approx 1 μ s positive pulse that has leading edge coincident with Q-Switch activation.

External triggering timing charts are shown in Figures 13 and 14. There are two modes available – single pulse of fixed pulse duration (EXTERNAL SP mode) or two independent separate pulses (EXTERNAL TP mode).



10.1.2 External Single-Pulse Triggering

In EXTERNAL SP triggering mode single pulse should be applied to LD TRIG socket with following parameters:

Exact triggering pulse duration can be found in laser *Test_data* file (Can be downloaded from HELP tab).

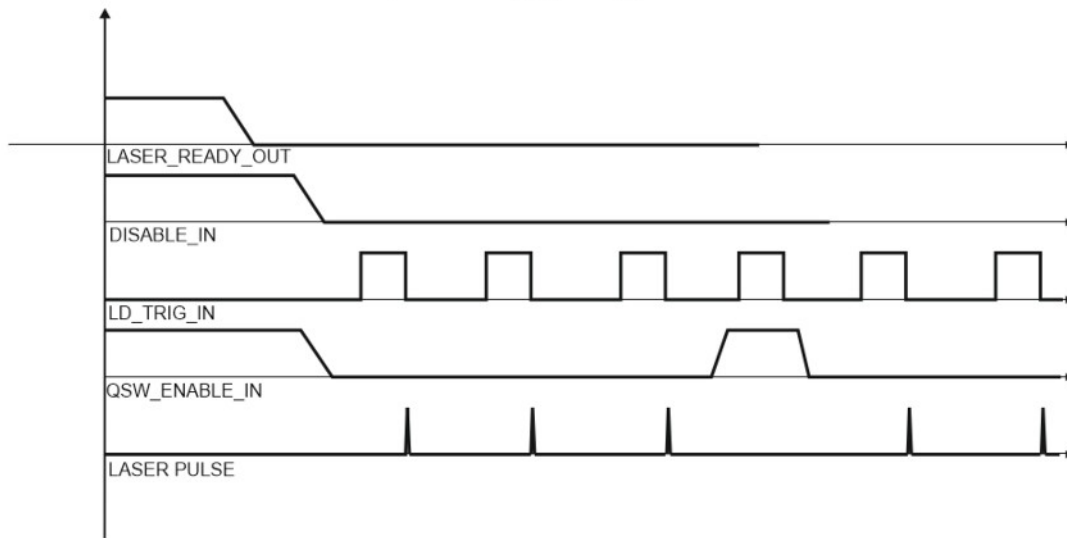
Pulse duration: $445 \pm 5 \mu\text{s}$ for Nd:YLF (1053 nm) laser;
Pulse duration: $300 \pm 5 \mu\text{s}$ or $208 \pm 5 \mu\text{s}$ for Nd:YAG (1064 nm) laser;
Minimum amplitude: 3V (for delay generator with 50 Ohm output impedance);
Input Voltage (Max): 5.3V;
Input Voltage (Min): -0.2V;
Positive-going threshold voltage (Max): 1.5V
Negative-going threshold voltage (Min): 0.65V

Pump diode is activated by rising edge of pulse, while Q-switch is activated by falling edge of input pulse. In order for laser to operate properly, the triggering pulse duration and pulse repetition rate should be as specified in [chapter 16](#).

QSW ENB input can be used for disabling Q-Switch activation. High level voltage, applied to this input before applying LD TRIG pulse will inhibit Q-switch and no laser pulse will be produced. Timing diagram of single pulse mode is presented in Figure 14 below.



Laser timing diagrams (external) 1-pulse triggering method



Laser pump cycle in detail

Note: Time axis scale is variable for clearer presentation
Arrows shows causal relationship

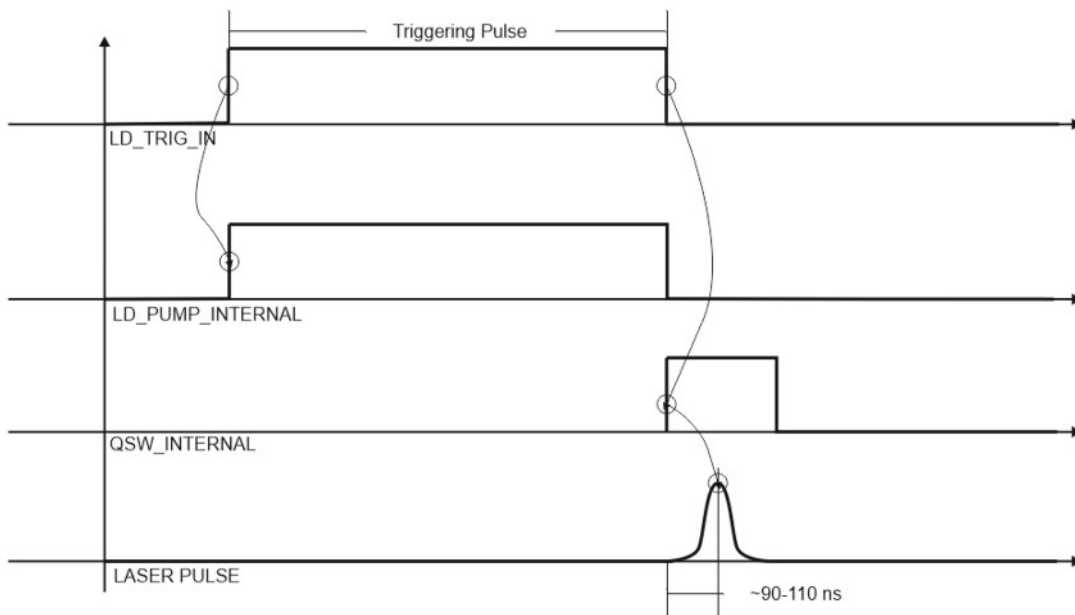


Figure 14: Timing chart of Q2 laser in EXTERNAL SP triggering mode



10.1.3 External Two-Pulse Triggering

Note:

External TP mode is not available in Q2 laser model by default. Please contact QLI for External TP mode operational details.

In EXTERNAL TP triggering mode separate triggering pulses for pump diode and Q-switch should be applied to LD TRIG and QSW ENB sockets. Pulse parameters should be following:

Pulse duration: $20 \pm 10 \mu\text{s}$;
Minimum amplitude: 3V (for delay generator with 50 Ohm output impedance);
Rise time: 100 ns max.
Input Voltage (Max): 5.3V;
Input Voltage (Min): -0.2V;
Positive-going threshold voltage (Max): 1.5V
Negative-going threshold voltage (Min): 0.65V

Temporal separation between two triggering pulses should be in 400-460 μs range for Nd:YLF laser (240-310 μs for Nd:YAG type laser). Pump diode and Q-Switch will be triggered from rising edges of pulses. Shorter delay of QSW ENB pulse will result in lower laser pulse energy due shorter pump diode pulse.

There is difference in control logic for Q1 and Q2/Q2HE/Q-SPART/Q-TUNE lasers:

For Q1 laser, if QSW ENB pulse is applied not in the required range, the laser pulse will be produced with fixed delay: $\sim 450 \mu\text{s}$ for Nd:YLF laser and $\sim 310 \mu\text{s}$ for Nd:YAG laser.

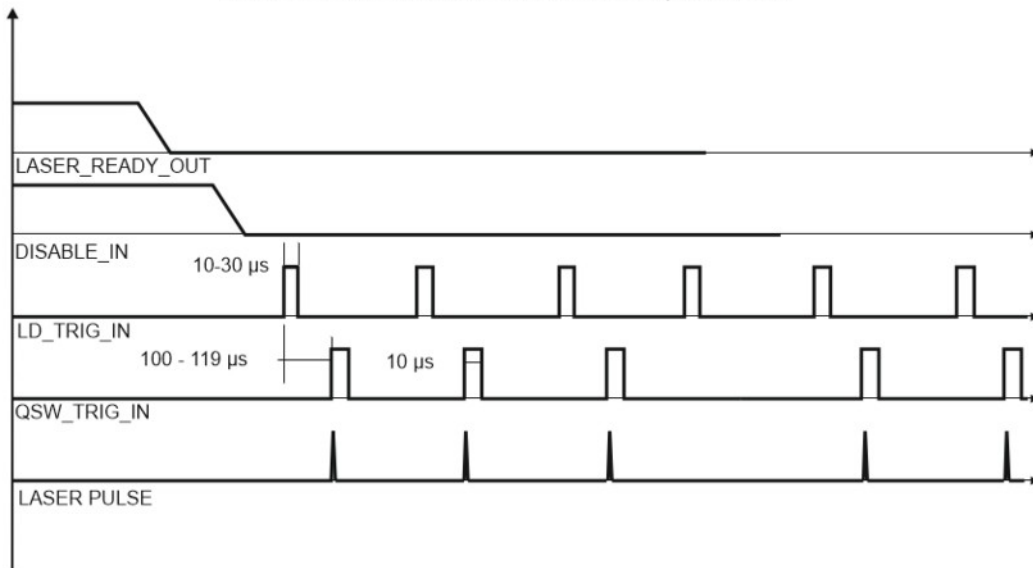
For Q2/Q2HE/Q-SPARK/Q-TUNE lasers, if QSW ENB pulse is applied not in the required range, it will be ignored and laser pulse will not be produced.

Timing diagram of two-pulse mode is presented in Figure 14 below.



Laser timing diagrams (external) 2-pulse triggering method

Note: Time axis scale is variable for clearer presentation



Laser pump cycle in detail

Note: Time axis scale is variable for clearer presentation
Arrows shows causal relationship

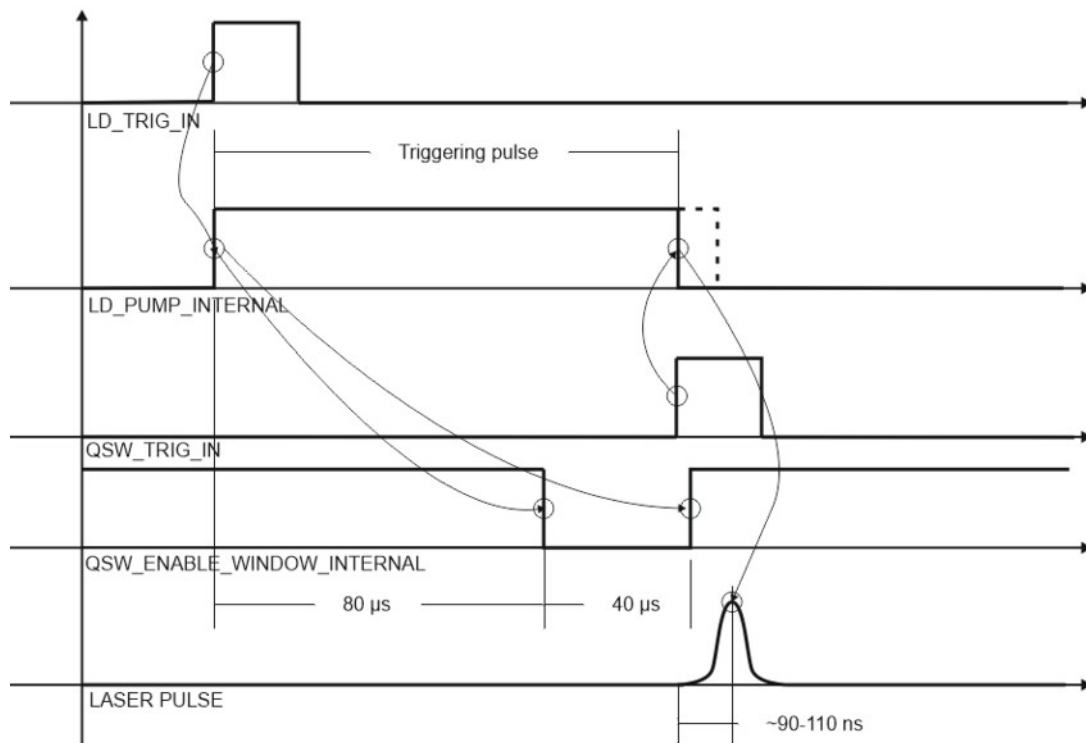


Figure 15: Timing chart of Q2 laser in EXTERNAL TP triggering mode



10.2 Turning on and Operating the Laser in Internal Triggering Mode

1. Turn key-switch to position ON. The STATUS LED should change the colour from green to red.
2. Laser will start initialization procedure which takes approx 15 seconds. Once initialization procedure is completed STATUS LED will change to yellow (or green blinking, which indicates that laser is warming up).
3. Press SHUTTER button. STATUS LED should start blinking yellow and SHUTTER button should be illuminated with yellow ring.

Note:

Radiation parameters of the laser stabilize after about several minutes warm-up time.

10.3 Laser Operation in External Triggering Mode

1. External triggering mode can be enabled from web control interface. Set Triggering Mode to “External SP” or “External TP” using “Toggle trig mode” button.
2. Triggering pulses should be applied as described in this Chapter above.

Note:

Failing to set correct triggering pulse parameters might lead to damage of the laser. It is user responsibility to make sure that triggering pulse has parameters as specified in this chapter.

3. Press SHUTTER button on the front panel of controller to enable laser firing.



10.4 Changing Laser Pulse Energy.

10.4.1 Changing Laser Pulse Energy by Change of Pump Current

1. Follow laser safety requirements as described in [chapter 2.5](#) at all times.
2. It is recommended to use laser at minimum pulse energy level, required for your task.
3. Pump current can be changed from “Service” tab as shown in Figure 16.

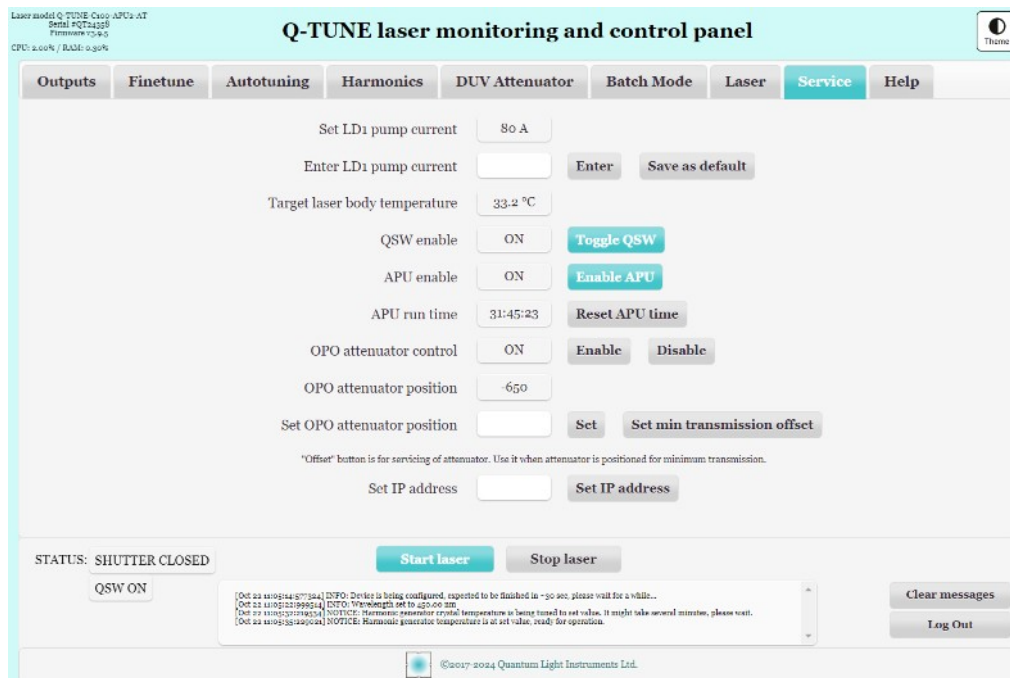


Figure 16: Change of pulse energy by adjusting pump current.

4. Typical laser generation threshold is in 70-80 A range, pump current corresponding to maximum pulse energy can be find in *Test_Data* file. Depending on required pulse energy, set appropriate pump current level in “Set LD1 pump current” field. Push “Enter” button once you entered required value. To make pump current value permanent, push “Save as default” button.
5. It is recommended to use laser at minimum pulse energy level, required for your task.

10.4.2 Changing OPO and 3rd Harmonic Pulse Energy by build-in FF Bypass

1. Laser pulse energy can be changed by setting “FF bypass ratio” value from “Outputs” tab as shown in Figure 17. 0% bypass corresponds to maximum pulse energy of OPO and 3rd harmonic. Required bypass value can be set in “Set bypass ratio” field. Push Enter button on keyboard or click outside input field area for executing the command.



Laser model: Q-TUNE-C400-APU's-AT
Serial: QTT1338
Firmware: V2.0.0
CPU: 2.09% / RAM: 0.36%

Q-TUNE laser monitoring and control panel

Theme

Outputs
Finetune
Autotuning
Harmonics
DUV Attenuator
Batch Mode
Laser
Service
Help

Port 1 is for tunable wavelength OPO beam.

Port #1 status: Inactive Enable OPO

Port #1 wavelength: 450 nm

Set wavelength: Set

Port 2 is for fundamental or third harmonic beam. When Port #2 is active, OPO is disabled.

Port #2 status: Inactive Max FF output Max TH output

Port #2 wavelength: 1064.2 nm

Simultaneous fundamental frequency (FF) output level from Port 2 can be smoothly adjusted by changing bypass ratio here.

FF bypass ratio: 0.00 %

Set bypass ratio: ▲ ▼

Port 3 is for SH beam. When OPO is operating residual SH beam is produced.

Port #3 status: Max Max SH output

Port #3 wavelength: 532.1 nm

Triggering mode: Internal Toggle trig modes Save as default

Pulse repetition rate: 100 Hz

Divide repetition rate by: Set divide ratio Save as default

STATUS: SHUTTER CLOSED

QSW ON

Start laser Stop laser

Clear messages

Log Out

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Figure 17: Change pulse energy by build-in FF bypass

10.4.3 Changing OPO pulse energy using optional attenuator

1. If your laser has **optional** build-in or attachable OPO attenuator pulse energy can be changed with the help of it. Using attenuator is recommended when it is necessary to keep constant pulse duration, pulse-to-pulse stability and beam divergence and change pulse energy only.
2. Pulse energy can be changed by changing the transmission (0 – 100 %) of the attenuator in laser user interface “OPO Attenuator” window (Figure 18).
3. Enter required value of transmission in “Set Transmission in %” field and push Enter button or click outside of the input field. The message informing about changed transmission value should appear in user message area.



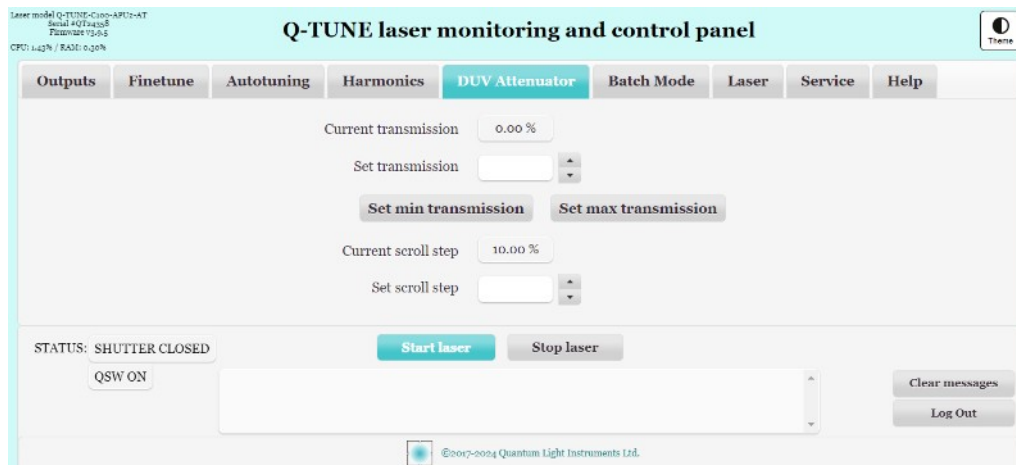


Figure 18: Change of laser output pulse energy by **optional** attachable attenuator

10.5 Temporarily Stopping Laser Operation

1. Press SHUTTER button on the front panel of controller. Press SHUTTER button again to resume system operation.
2. Alternatively, laser operation can be stopped by pushing “STOP LASER” button from laser web-interface (see Chapter [Chapter 11](#) for details).

10.6 Turning Laser off

1. Release SHUTTER button on controller to off position.
2. Turn keyswitch to OFF position

Note:

If optional harmonics module is supplied, do not disconnect the power supply or APU unit from the mains. The harmonics crystals should be constantly heated.

10.7 Laser Interlocks

QLI lasers have two interlock inputs: **DISABLE** and **E-STOP**. DISABLE interlock is non-latching, this input can be used to interlock laser for repetitive laser start – stop operation. E-Stop interlock input is latching, meaning that laser electrical supply must be cut off in order to reset this interlock.

Active interlock means laser emission is disabled. Both inputs are active high: leave interlock signal line unconnected (or apply 3.5 V .. 5.0V) in order to activate interlock. Electrically short signal line to ground (or apply 0V .. 1.5V) in order to deactivate interlock.

Access points to interlock signals can differ, depending on laser model. Most lasers have interlock BNC connectors mounted on the rear panel of the APU unit (a), others have interlock free-hanging (pigtail) interlock BNC connectors as part of cable assembly (b), see “Figure 18. Possible “Interlocks” configurations”.



Interlock lines are delivered electrically pre-shorted out of the factory, by including appropriate interlocking connectors, thus enabling laser emission out of the box at user site. 0 to 75 Ohm BNC terminators are included both variants (a) or (b).

It is worth to mention, that all interlock signal lines are electrically identical regardless of interlock access location.

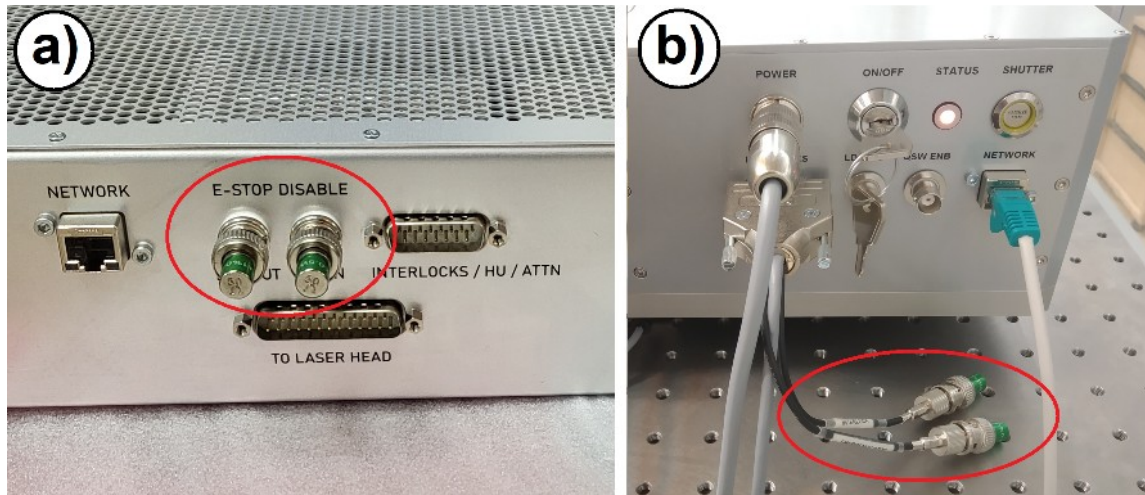


Figure 19: Possible “Interlocks” configurations: a) BNC connectors on APU unit b) Pig tail BNC connectors on laser back panel.



11 Remote Control

11.1 Connecting to laser head

The remote control can be done from a personal computer through NETWORK (RJ-45) socket on the front panel of the controller.

Up to 250 systems can be controlled from single computer (determined by available network address space).

Ethernet interface can be configured in two ways. For settings please refer to *Test_Data* file.

Static IP address

For direct Q2-to-computer connection, each controller has assigned static IP address, provided in *Test_Data* file.

Configure your computer ethernet adapter for communication with Q2 laser. For this you have to disable automatic IP address assignment and set computer IP address as 192.168.1.xx, where xx any number in range 1-254 (address should be different to laser IP address presented in *Test_Data* file).

Use crossover cable for communication directly with PC. The difference between straight and crossover cable is shown in Figure 20. Note, for Q2 to router connection (described in next chapter) straight cable is necessary.

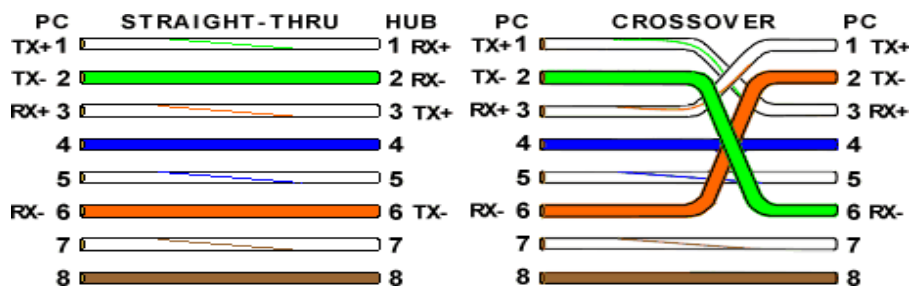


Figure 20: Straight-thru vs crossover ethernet cable pinout



In Windows 10 static IP address of your computer can be configured in following way:

1. Push Start button (left bottom corner of screen) and choose “Settings”. In Settings window, choose “Network and Internet”. Choose “Ethernet” and “Change adapter settings”.

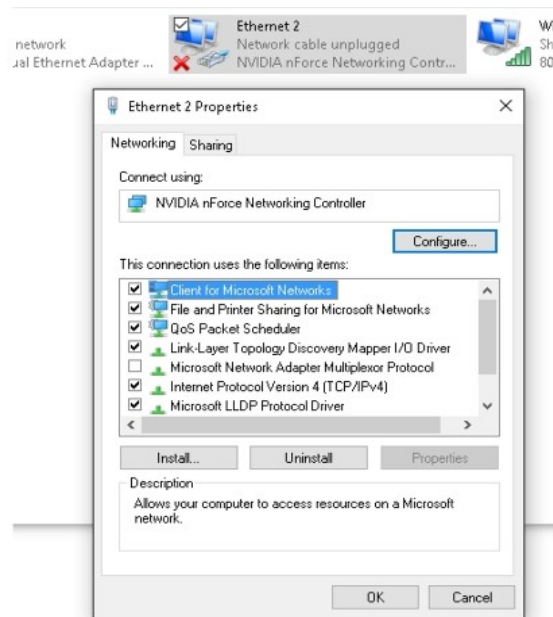


Figure 21: Ethernet 2 properties

2. The list of available Ethernet adapters will open. Right click on your Ethernet adapter and choose “Properties”. The window for change adapter properties will open. Select “Internet protocol Version 4 (TCP/IPv4)” and click “Properties”.
3. Select “Use following IP address” and enter your computer IP address and Subnet mask as shown in picture above. Press “OK” button.

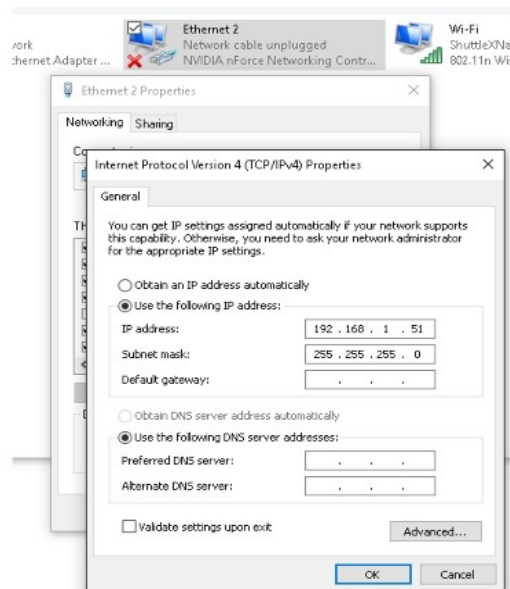


Figure 22: Internet protocol (TCP/IPv4) properties



4. In web-browser address field please enter:
<http://192.168.1.xx:7557>, (check [chapter 16](#) for actual address).

You will be directed to login screen:

Enter user name and password as found in [chapter 16](#).

If login is successful, you will be redirected to the main Q2 control screen.



11.2 Outputs Tab

The screenshot shows the 'Outputs' tab of the Q-TUNE laser monitoring and control panel. The panel has a light blue header with the title 'Q-TUNE laser monitoring and control panel' and a 'Theme' button. Below the header is a navigation bar with tabs: 'Outputs' (selected), 'Finetune', 'Autotuning', 'Harmonics', 'DUV Attenuator', 'Batch Mode', 'Laser', 'Service', and 'Help'. The main content area is divided into sections for three ports. Port #1 is for a tunable wavelength OPO beam. It shows 'Port #1 status' as 'Inactive' with an 'Enable OPO' button. 'Port #1 wavelength' is set to '450 nm' with a 'Set wavelength' input field and a 'Set' button. Port #2 is for fundamental or third harmonic beam. It shows 'Port #2 status' as 'Inactive' with 'Max FF output' and 'Max TH output' buttons. 'Port #2 wavelength' is set to '1064.2 nm'. Below this is a section for 'Simultaneous fundamental frequency (FF) output level from Port 2' with a 'FF bypass ratio' of '0.00 %' and a 'Set bypass ratio' input field with up/down arrows. Port #3 is for SH beam. It shows 'Port #3 status' as 'Max' with a 'Max SH output' button. 'Port #3 wavelength' is set to '532.1 nm'. At the bottom of the main area are 'Triggering mode' (Internal), 'Pulse repetition rate' (100 Hz), and 'Divide repetition rate by' (input field) with 'Toggle trig modes', 'Save as default', 'Set divide ratio', and 'Save as default' buttons. The bottom status bar shows 'STATUS: SHUTTER CLOSED', 'QSW ON' button, 'Start laser' and 'Stop laser' buttons, a message log with 'Clear messages' and 'Log Out' buttons, and a copyright notice '©2017-2024 Quantum Light Instruments Ltd.'.

Figure 23: Outputs tab of Q-Tune control panel

Outputs window should be used for setting output wavelength of OPO, activating or deactivating auxiliary laser ports.

STATUS field shows current status of the laser. Possible values are “SHUTTER CLOSED”, “RUNNING”, “WARMING UP”, “WAITING FOR PC COMMAND”, “WAITING FOR EXTERNAL TRIGGER”.

The user can start or stop the laser (SHUTTER button on front panel of controller should be in ON position) by pushing “Start laser” or “Stop laser” buttons.

Port #1 is for OPO beam. In order to access OPO radiation, check “Port #1” status. It should be “Max”. If not, push “Enable OPO” button. Button state should change to “OPO Enabled”.

Wavelength of OPO can be changed by entering required value in “Set wavelength” field and pushing “Set” button.

Port #2 can be configured for third or fundamental harmonics outputs. For fundamental harmonic output, push “Max FF output” button. For third harmonic output push “Max TH output” button. OPO output is not available when Port #2 is activated.

Port #3 provides access to the second harmonic beam of pump laser. Residual SH beam is available when OPO is enabled or when third harmonic output is enabled at the port #2. Pulse energy is lower since part of SH beam is up-converted to the third harmonic beam. For maximising second harmonic output, push “Max SH



output” button. The third harmonic crystal will be detuned and maximum SH pulse energy produced.

The user can change triggering mode of the laser. By default the triggering mode is set to “Internal”, when laser pulse repetition rate is set by build-in delay generator. By pushing “Toggle trig mode” button the triggering mode can be changed to “External SP” or “External TP” - the laser will be triggered by pulse applied to LD TRIG socket on the front panel of controller. (See required triggering pulse parameters in [Chapter 10](#)).

Current triggering mode can be saved to the non-volatile memory by pushing button “Save as default”.

It is possible to divide laser pulse repetition rate by integer number from 1 to 100. For this, enter divide value into “Divide repetition rate by” fields and push button “Set divide ratio”. This button store value in volatile memory.

It is possible to save new repetition rate to non-volatile memory by pushing button “Save as default”.



11.3 Finetune Tab

Laser model Q-TUNE-C100-APU2-V7
Serial #QT121395
Firmware 1.3.0.5
CPU: 1.55% / RAM: 0.30%

Q-TUNE laser monitoring and control panel

Theme

Outputs **Finetune** Autotuning Harmonics DUV Attenuator Batch Mode Laser Service

Help

Wavelength 450 nm

Set wavelength Set

Calculated OPO signal wavelength for monitoring or calibration.

Monitor wavelength 450 nm

Finetune OPO wavelength in 210-2300 nm range and save it to the non-volatile memory.

OPO wavelength deviation 22

Inc/Dec deviation Save as default

Finetune SHG output energy in 210 - 409.9 nm range and save it to the non-volatile memory. NOTE: this control does not have effect in 410-2300 nm range.

SHG energy deviation 30

Inc/Dec deviation Save as default

Finetune beam pointing and save it to the non-volatile memory.

Pointing deviation -2

Inc/Dec deviation Save as default

Finetune OPO phase and save it to the non-volatile memory.

Phase deviation -24

Inc/Dec deviation Save as default

Use this function when OPO wavelength and pulse energy is calibrated/optimized at 532.1 nm.

Save OPO look-up table offset Save

Use this function when SHG pulse energy is calibrated/optimized at 222.4 or 297.2 nm wavelengths.

Save SHG look-up table offset Save

Use this function when OPO wavelength and pulse energy is calibrated/optimized at 532.1 nm.

Save Pointing look-up table offset Save

Use this function when OPO wavelength and pulse energy is calibrated/optimized at 532.1 nm.

STATUS: SHUTTER CLOSED Start laser Stop laser

QSW ON

Clear messages Log Out

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Figure 24: Finetune tab of Q-Tune control panel

Finetune window is provided for correction of OPO tuning curve and pulse energy. Due stepper motor rotation non-linearities or environmental factors the position of non-linear crystals in OPO cavity might deviate for optimal one. Finetune window controls helps to restore optimal positions.

“Energy position deviation” and “Wavelength position deviation” fields indicates deviations for factory-set OPO crystals positions. The units in these windows are indicated in microsteps of stepper motors.

“Inc/Dec deviation” field adds or reduces deviation by entered value. For sending command, please push “Enter” button on keyboard. Note, “Save as default” command does not send value to the laser, it saves current deviation in the non-volatile memory.

The procedure of calibrating OPO wavelength at required wavelength is following:

1. Enter wavelength of interest in “Set wavelength” input field. Push “Set” button in order to send command to the laser. Currently entered wavelength calibration deviation is indicated in “Wavelength position deviation” field.



2. While monitoring OPO wavelength with external spectrometer or spectrograph, enter +1.5 or -1.5 in “Inc/Dec deviation” field and push “Enter” on keyboard. The field “Wavelength position deviation” should change, indicating that laser received and executed the commands.

3. Check if beam wavelength changed. Repeat “Inc/Dec deviation” command several times in order to match OPO produced wavelength with entered in the step 1. Actual current deviation number should be monitored by the field “Wavelength position deviation”.

4. Once optimal value is found, it can be stored into non-volatile memory by pushing “Save as default” button.

The field “Save look-up table offset” is used for calibration of OPO wavelength at second harmonic wavelength of pump laser (526.5 or 532.1 nm, depending on pump laser used). It offsets the whole factory-set tuning curve by the values indicated in “Energy position deviation” and “Wavelength position deviation”. Use this command for initial unit calibration or if it is detected that there is uniform pulse energy position or wavelength position deviation across whole tuning range.

“Pointing deviation” field indicates deviation of spectral selector prism from factory-set position. The units is indicated in microsteps of stepper motors.

Due material properties change over wide wavelength range, it might be necessary to correct pointing deviation. Change of deviation results in beam direction change in vertical plane.

The procedure of calibrating OPO pointing at required wavelength is following:

1. Enter wavelength of interest in “Set wavelength” input field. Push “Set” button in order to send command to the laser. Currently entered wavelength calibration deviation is indicated in “Pointing deviation” field.

2. Monitor beam position at the output of OPO using beam profiler or target for beam visualisation. Enter +1.5 or -1.5 in “Inc/Dec deviation” field and push “Enter” on keyboard or click outside input field. The field “Pointing deviation” should change, indicating that laser received and executed the commands.

3. Check if the beam position changed. Repeat “Inc/Dec deviation” command several times in order to adjust beam pointing as needed. Current deviation number should be monitored by the field “Pointing deviation”.

4. Once optimal value is found, it can be stored into non-volatile memory by pushing “Save as default” button.

The field “Save PRISM look-up table offset” is used for calibration of beam pointing at second harmonic wavelength of pump laser (526.5 or 532.1 nm, depending on pump laser used). It offsets the whole factory-set tuning curve by the value indicated in “Pointing deviation”. Use this command for initial unit calibration or if it is detected that there is uniform pointing deviation across whole tuning range.

“Phase deviation” field indicates deviation of phase corrector plate from factory-set position. The units is indicated in microsteps of stepper motors.

Due material properties change over wide wavelength range, it might be necessary to phase deviation for best OPO beam spatial profile. Change of phase deviation results in beam profile change.



The procedure of calibrating OPO phase at required wavelength is following:

1. Enter wavelength of interest in “Set wavelength” input field. Push “Set” button in order to send command to the laser. Currently entered wavelength calibration deviation is indicated in “Phase deviation” field.
2. Monitor beam profile at the output of OPO using beam profiler or white screen for beam visualisation. Enter +1..5 or -1..5 in “Inc/Dec deviation” field and push “Enter” on keyboard or click outside input field. The field “Phase deviation” should change, indicating that laser received and executed the commands.
3. Check if the beam position changed. Repeat “Inc/Dec deviation” command several times in order to adjust beam pointing as needed. Current deviation number should be monitored by the field “Phase deviation”.
4. Once optimal value is found, it can be stored into non-volatile memory by pushing “Save as default” button.

The field “Save phase look-up table offset” is used for calibration of OPO phase at default wavelength, depending on model it could be 532.1 nm or 690 nm. It offsets the whole factory-set tuning curve by the value indicated in “Phase deviation”. Use this command for initial unit calibration or if it is detected that there is uniform phase position deviation across whole tuning range.



11.4 Attenuator Tab

When supplied with OPO attenuator, extra tab will appear allowing transmission control of the attenuator.

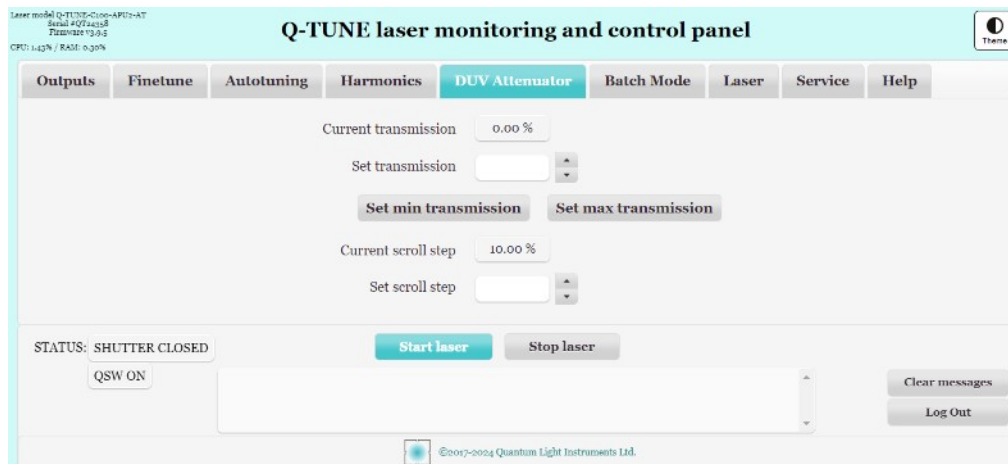


Figure 25: OPO attenuator tab of Q-Tune control panel

The window allows the user control attenuator transmission. Enter required value of transmission in “Set Transmission in %” field and push Enter button or click outside of the input field. The message informing about changed transmission value should appear in user message area.

In “Set Scroll Step” input field granularity of transmission value can be changed with the use of up or down buttons.



11.5 Autotuning Tab

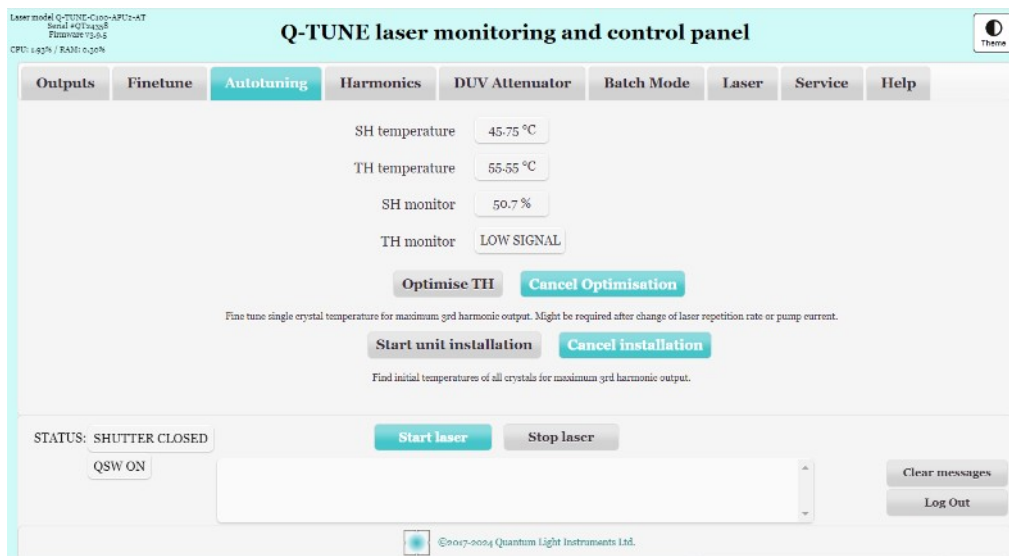


Figure 26: Autotuning tab of Q-Tune control panel

Autotuning window allows automatically find crystal temperatures, depending on pulse energy and/or pulse repetition rate.

Several help functions are available:

“Start unit installation” button starts search of all crystals temperatures that produces maximum output at each of the harmonics wavelength. This function should be used at initial setup of the unit, or when it was found that both second and third harmonic output is much lower than specified value.

This search typically allows to find crystal temperatures that are approximately optimal for each of harmonic wavelength if taken separately. However, crystal temperatures are inter-dependent. For example, for maximum third harmonic output second harmonic crystal temperature should be slightly detuned from its max value for 532.1 nm generation.

“Optimise TH” buttons allows to fine tune each individual crystal temperature for maximum of specified output wavelength. These buttons should be used if lower than specified in the datasheet harmonic output is detected.

The automatic tuning functions should be used after laser warm-up and under settings (pump repetition rate and pump current) that will be used. The laser should be firing when automatic tuning is activated, otherwise an error message will be produced.

During the scan it is possible to monitor crystal temperatures and readings of energy monitors. The monitor values should be in 10%-90% range for optimal operation.



11.6 Harmonics Tab

The screenshot shows the 'Harmonics' tab of the Q-TUNE laser monitoring and control panel. The panel has a light blue header with the title 'Q-TUNE laser monitoring and control panel' and a 'Theme' button. Below the header is a navigation bar with tabs: Outputs, Finetune, Autotuning, Harmonics (selected), DUV Attenuator, Batch Mode, Laser, Service, and Help. The main content area displays various parameters and controls for the TH/OPO-SHG-GP module. It includes a note about manually changing harmonic generator crystal temperatures. Parameters shown include SH monitor (50.7%), TH monitor (0%), SH temperature (45.75 °C), Set SH temperature (input field), TH temperature (55.54 °C), Set TH temperature (input field), OPO temperature (33.98 °C), Set OPO temperature (input field), OPO-SHG temperature (34.01 °C), Set OPO-SHG temperature (input field), Pulse counter (9989250), On time (204:49:27), Environment humidity (10.8 %), Environment pressure (1004.06 mBar), and CPU temperature (54.62 °C). Each temperature parameter has a 'Set' button and a 'Save as default' button. At the bottom, there is a status bar showing 'STATUS: SHUTTER CLOSED', a 'QSW ON' button, 'Start laser' and 'Stop laser' buttons, a 'Clear messages' button, and a 'Log Out' button. A copyright notice '©2017-2024 Quantum Light Instruments Ltd.' is visible at the very bottom.

Parameter	Value	Control
Module type	TH/OPO-SHG-GP	
Manually change harmonic generator crystal temperatures for optimal performance.		
SH monitor	50.7 %	
TH monitor	0 %	
SH temperature	45.75 °C	
Set SH temperature		Set Save as default
TH temperature	55.54 °C	
Set TH temperature		Set Save as default
OPO temperature	33.98 °C	
Set OPO temperature		Set Save as default
OPO-SHG temperature	34.01 °C	
Set OPO-SHG temperature		Set Save as default
Pulse counter	9989250	
On time	204:49:27	
Environment humidity	10.8 %	
Environment pressure	1004.06 mBar	
CPU temperature	54.62 °C	

STATUS: SHUTTER CLOSED

QSW ON

Start laser Stop laser

Clear messages Log Out

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Figure 27: Harmonics tab of Q-Tune control panel

If your laser has harmonics unit, switching between harmonics can be done in “Harmonics” tab. For desired harmonic press “Max XX output” button. Wait several minutes before crystal temperature is set.

Manual fine tuning of crystal temperatures can be done in this tab. Use external power or energy meter to observe energy change. Enter temperature value 0.1 degree higher or lower than current in the “Set temperature” field for selected harmonic. If output energy increases, repeat previous step. If energy decreases, change temperature in the opposite direction. Continue until maximum energy is reached.

“Save as default” buttons save entered value to non-volatile memory, saved value will be used as default value after next power up. Default temperatures can not be updated when module is in installation or optimization mode.

Ambient humidity and pressure sensors might be installed in harmonics unit, current values are displayed in this tab.



11.7 Laser Tab

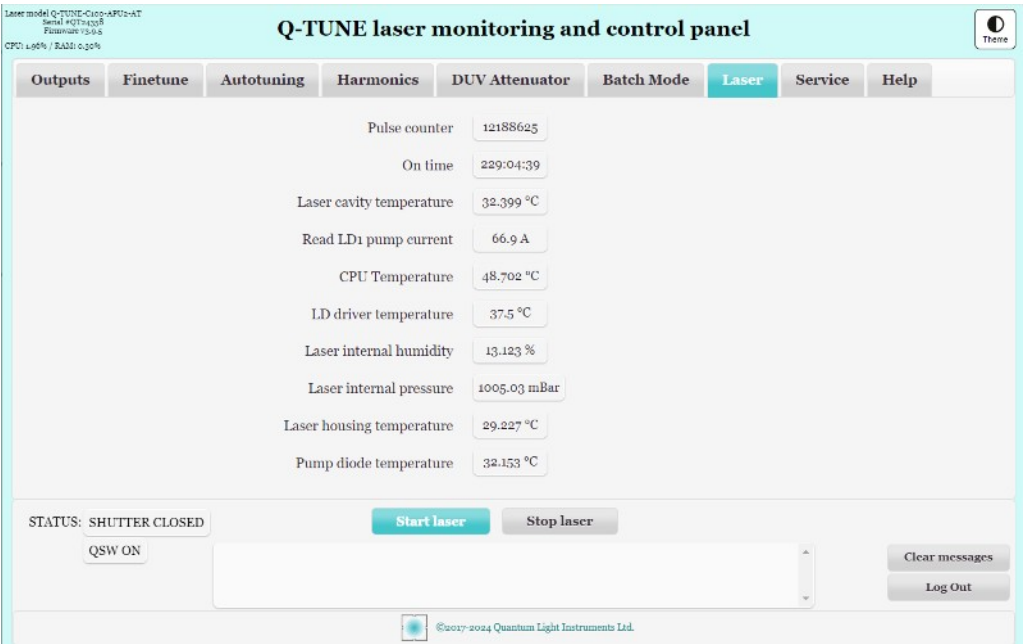


Figure 28: Laser tab of Q-Tune control panel

Main pump laser parameters, like pump current, laser cavity and housing temperatures, number of pulses made, triggering mode and pulse repetition rate, humidity and pressure inside the laser head can be monitored on this tab.

The user can start or stop laser (SHUTTER button on front panel of controller should be in ON position) by pushing “Start laser” or “Stop laser” buttons.



11.8 Batch mode Tab

The screenshot shows the 'Batch Mode' tab of the Q-TUNE laser monitoring and control panel. The interface includes a top navigation bar with tabs: Outputs, Finetune, Autotuning, Harmonics, DUV Attenuator, Batch Mode (selected), Laser, Service, and Help. The main area is titled 'Configure OPO scan parameters: scan range, scan step, pulses made at each wavelength.' It contains several input fields: 'Scan start wavelength', 'Scan end wavelength', 'Wavelength scan step', 'Pulses per step', and 'Repeat scan'. A 'Configure Scan' button is located next to the 'Repeat scan' field. Below these fields, a status message reads: 'Scan is configured with following parameters: Start wavelength: 410.0 nm, End wavelength: 700.0 nm, Step: 10.0 nm, Pulses per step: 100, Repeats: 0'. There are two buttons: 'Start Scan' and 'Stop Scan'. Below these, the 'Scan status' is shown as 'Idle'. The 'Current wavelength' is '450 nm' and 'Pulses remaining' is '0'. The 'Batch Mode' is set to 'Batch Off' with a dropdown arrow. The 'Number of pulses in batch' is '0' and the 'Currently set pulses in batch' is '1500000'. A 'Set batch' button is next to the 'Number of pulses in batch' field. At the bottom, there is a 'STATUS: SHUTTER CLOSED' indicator, a 'QSW ON' button, and 'Start laser' and 'Stop laser' buttons. A 'Clear messages' button and a 'Log Out' button are also present. The footer shows a copyright notice: '©2017-2024 Quantum Light Instruments Ltd.'.

Figure 29: Batch Mode tab of Q-Tune control panel

In Batch Mode window user can configure automatic scanning of the wavelength.

Scan start and finish wavelength in nm, scan step in nm, pulses to be made at each wavelength during scan and number of scans to be made can be entered into respective input fields. Once all settings are configured, push “Configure Scan” button. If scanning settings are valid, the message will appear with configured values.

In order to start scan, push “Start Scan” button. To stop scan, push “Stop Scan” button. After “Start Scan” button is pressed, laser will operate in Q-Switch off mode for 10 seconds, then preset number of pulses will be delivered. “Stop Scan” button will reset number of pulses made and stop laser operation.

Batch mode allows to produce preset number of pulses. Batch mode can run in two configurations:

“Single Batch” a set of pulses fired from laser, with pumping diodes turned off after the batch is finished.

“Continuous Batch” a set of pulses fired from laser, with pumping diodes continuously running after the batch is finished.

Once Batch mode is set-up, pushing “Start laser” button will produce batch of preset number of pulses. Laser firing will automatically stop after batch execution is completed.



11.9 Service Tab

The screenshot displays the 'Service' tab of the Q-TUNE laser monitoring and control panel. The top navigation bar includes tabs for Outputs, Finetune, Autotuning, Harmonics, DUV Attenuator, Batch Mode, Laser, Service (selected), and Help. The main content area contains several control fields and buttons:

- Set LD1 pump current:** A text input field showing '80 A' and a 'Set' button.
- Enter LD1 pump current:** A text input field with an 'Enter' button and a 'Save as default' button.
- Target laser body temperature:** A text input field showing '33.2 °C'.
- QSW enable:** A dropdown menu set to 'ON' and a 'Toggle QSW' button.
- APU enable:** A dropdown menu set to 'ON' and an 'Enable APU' button.
- APU run time:** A text input field showing '31:45:23' and a 'Reset APU time' button.
- OPO attenuator control:** A dropdown menu set to 'ON' and 'Enable'/'Disable' buttons.
- OPO attenuator position:** A text input field showing '-650'.
- Set OPO attenuator position:** A 'Set' button and a 'Set min transmission offset' button.
- Set IP address:** A text input field and a 'Set IP address' button.

At the bottom, there is a 'STATUS: SHUTTER CLOSED' indicator, a 'QSW ON' indicator, and 'Start laser'/'Stop laser' buttons. A message log at the bottom right shows system messages, including a warning about device configuration and a notice about harmonic generator temperature. The footer includes the copyright notice '©2017-2024 Quantum Light Instruments Ltd.'.

Figure 30: Service tab of Q-Tune control panel

For setting pump current, enter numerical value into “Set LD1 pump current window” and push “Set Current” button. If you want to make this value permanent, push “Save as default” button.

The field “Target laser body temperature” indicates the temperature that laser body needs to reach for specified performance. This temperature is reached automatically by installed heaters and/or coolers.

If laser is configured with E-O Q-Switch, extra fields will appear for control of Q-Switch.

“Toggle QSW” button allows to stop laser pulse emission while pump diodes are still operating. This allows to keep thermal lens inside lasing rods constant, so after re-activation of the Q-Switch the first pulse will be with the same energy and divergence as the following ones.

“APU Enable” fields indicates current status of Air-Purging Unit (APU). Unit can be enabled or disabled by toggling “Enable APU” button. The status of APU unit is saved to non-volatile memory.

“APU run time” field shows air purging unit pump operation time, reset the timer once new desiccant cartridge is installed.

Direct stepper motor controls are available if laser has **optional** attenuator or bypass module.

“OPO attenuator control” allows to disable motor if it is disconnected (otherwise error will be produced disabling laser operation). Also, motor can be disabled if its operation is not required (attenuation or bypass ratio should be fixed). Enable/disable setting is saved to non-volatile memory, motor will remain in disable state after powering cycle.



If necessary, stepper motor position can be changed manually from “Set OPO attenuator position” field. Current position of motor is indicated in “OPO attenuator position” field.

“Set min transmission offset” button should be executed if re-calibration of stepper motor tuning curve is required. **NOTE: This command should be executed when attenuator/bypass unit is configured for minimum transmission.**

“Set IP address” allows user to changed laser IP address. Last two numbers can be changed. For example 192.168.2.50

11.10 Help Tab

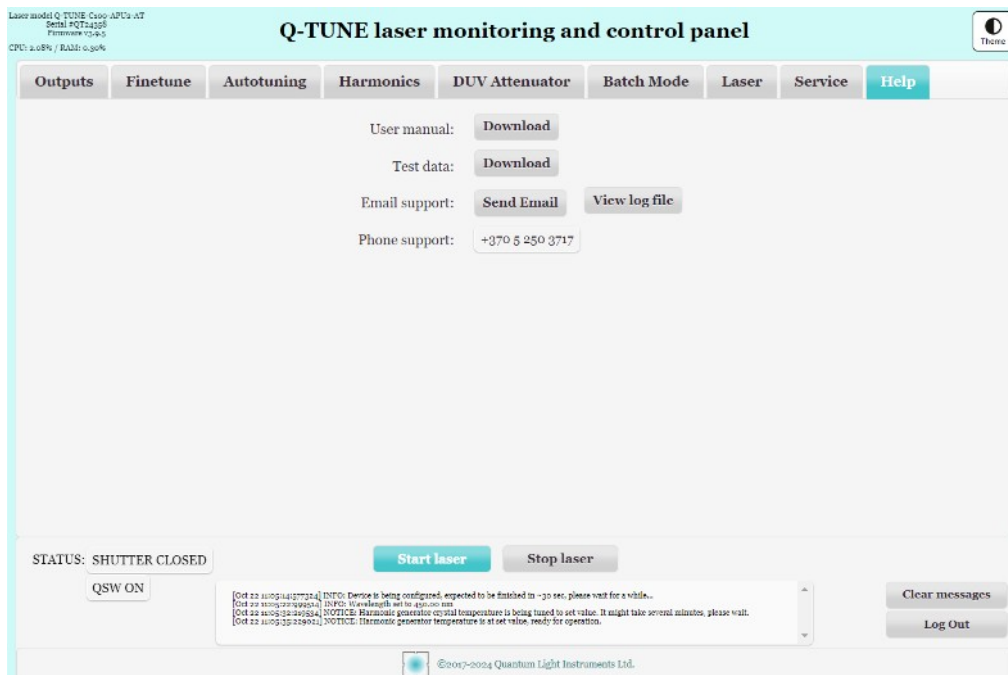


Figure 31: Help tab of Q-Tune control panel

From help window it is possible to download laser manual and test data, also manufacturers contact information (email address and phone) is provided.

Log files for troubleshooting are available in this tab. To acquire log file, press “View log file” button.



12 Programming

12.1 Programming Format

There are two special web-server pages that are intended for laser control from command line interface (using programs like cURL (<https://curl.haxx.se/>)) or even web-browser.

Data is sent to laser and received in JSON format.

1. The laser internal settings can be found on values page:

<http://192.168.1.xx:7557/values?> (xx value can be found in [Chapter 16](#)).

The laser will respond with HTTP message that contains HTTP header and data encapsulated into JSON array, containing various settings of the laser (each line is terminated by CR LF symbols (0x0D 0x0A in hex format)):

*HTTP/1.1 200 OK
Transfer-Encoding: chunked*

*22A
[{"device":"attenuator","values":{"attenuator":{"id":60,"type":0,"stepper-enable":1,"error-code":0,"transmission":50.1,"step-size":1.0,"zero-transmission-offset":230,"position":871}}},{ "device":"laser","values":{"id":"000009d1b15a","laser-run":1,"housing-temperature":27.750,"temperature":27.750,"center-temp":28.000,"pulse-count":977724,"error-code":0,"on-time":142255,"cpu-temperature":44.935,"LD1-current":69.8,"LD1-set-current":70.0,"trig-mode":0,"batch-pulses-set":0,"batch-mode":0,"batch-mode-pulses-remaining":0,"rep-rate":50.00,"laser-state":6}}]
0*

Data section starts with hexadecimal value of message length, and is finished with remaining bytes to be send (zero in example above) and is terminated with CR LF symbols (0x0D 0x0A in hex format)). Note, web browsers typically do not show HTTP headers and message length information.

Internal settings array is updated once per second, so it might take some time to get it updated once you send laser control command.

2. The laser control is done on send page:

http://192.168.1.xx:7557/send?JSON_OBJECT

JSON_OBJECT is URL encoded JSON object containing laser control command.

For example, in order to start laser shooting, the following command should be send with cURL:

curl -u username:password <http://192.168.1.xx:7557/send?%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22laser-run%22%3A%201%7D%7D%7D>

where username:password is user name and password that is provided in [chapter 16](#).



which is URL-encoded following message:

```
http://192.168.1.xx:7557/send?{"action":"control","code":  
{"device":"laser","values":{"laser-run":1}}}
```

When sending command from web-browser, command should be send without URL-encoding, since web-browser will do encoding automatically.

The laser will respond with message

```
HTTP/1.1 200 OK  
Transfer-Encoding: chunked
```

```
15  
{"message":"success"}  
0
```

if valid JSON object was sent, and

```
HTTP/1.1 200 OK  
Transfer-Encoding: chunked
```

```
14  
{"message":"failed"}  
0
```

if invalid JSON object was sent.

The laser response will not contain information whether the command was executed successfully or not, it only informs whether JSON object syntax is correct or not.

Whether command was executed or not the user can find on values page. After successful command execution, the corresponding internal setting will change.



12.2 List of Laser Commands (With and Without URL Encoding)

Table 6: List of Pump Laser Commands

Description	Command
Run laser	<pre>{ "action": "control", "code": { "device": "laser", "values": { "laser-run": 1 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22laser-run%22%3A%201%7D%7D%7D</pre>
Stop laser	<pre>{ "action": "control", "code": { "device": "laser", "values": { "laser-run": 0 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22laser-run%22%3A%200%7D%7D%7D</pre>
Set pulse repetition rate divide ratio	<pre>{ "action": "control", "code": { "device": "laser", "values": { "rep-rate-divide": X } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22rep-rate-divide%22%3A%20X%7D%7D%7D</pre> <i>X -should be integer number in 1-20 range</i>
Set default pulse repetition rate divide ratio	<pre>{ "action": "control", "code": { "device": "laser", "values": { "rep-rate-divide-default": X } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22rep-rate-divide-default%22%3A%20X%7D%7D%7D</pre> <i>X -should be integer number in 1-20 range</i>
Activate batch mode	<pre>{ "action": "control", "code": { "device": "laser", "values": { "batch-mode": 1 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22batch-mode%22%3A%201%7D%7D%7D</pre>
Deactivate batch mode	<pre>{ "action": "control", "code": { "device": "laser", "values": { "batch-mode": 0 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22batch-mode%22%3A%200%7D%7D%7D</pre>
Set number of pulses in batch	<pre>{ "action": "control", "code": { "device": "laser", "values": { "batch-pulses-set": X } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22batch-pulses-set%22%3A%20X%7D%7D%7D</pre> <i>X should be integer number in 1 - 10000 range</i>
Set pump current	<pre>{ "action": "control", "code": { "device": "laser", "values": { "LD1-current": 72.0 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22current%22%3A%2072.0%7D%7D%7D</pre>



	Current should be positive number in xxx.y format.
Save default pump current to non-volatile memory	<pre>{ "action": "control", "code": { "device": "laser", "values": { "LD1-current-default": 72.0 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22current-default%22%3A%2072.0%7D%7D%7D</pre> Current should be positive number in xxx.y format.
Set external SP (Single Pulse) triggering mode	<pre>{ "action": "control", "code": { "device": "laser", "values": { "trig-mode": 1 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22trig-mode%22%3A%201%7D%7D%7D</pre>
Set external TP (Two Pulse) triggering mode	<pre>{ "action": "control", "code": { "device": "laser", "values": { "trig-mode": 2 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22trig-mode%22%3A%202%7D%7D%7D</pre>
Set internal triggering mode	<pre>{ "action": "control", "code": { "device": "laser", "values": { "trig-mode": 0 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22trig-mode%22%3A%200%7D%7D%7D</pre>
Save triggering mode to non-volatile memory	<pre>{ "action": "control", "code": { "device": "laser", "values": { "trig-mode-default": X } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22trig-mode%22%3A%20X%7D%7D%7D</pre> Set X=0 for internal triggering and X=1 or X=2 for external triggering
Disable Q-Switch	<pre>{ "action": "control", "code": { "device": "laser", "values": { "qsw-run": 0 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22qsw-run%22%3A%200%7D%7D%7D</pre> No laser output is produced when Q-switch is disabled, pump diodes are running at preset repetition rate.
Enable Q-Switch	<pre>{ "action": "control", "code": { "device": "laser", "values": { "qsw-run": 1 } } }</pre> <pre>%7B%22action%22%3A%22control%22%2C%22code%22%3A%7B%22device%22%3A%22laser%22%2C%22values%22%3A%7B%22qsw-run%22%3A%201%7D%7D%7D</pre>
Set Q-switch divide mode	<pre>{ "action": "control", "code": { "device": "laser", "values": { "QSW-divide-mode": 1 } } }</pre> Set pump or Q-switch repetition rate divide mode. In the later case pump diodes operate at fixed repetition rate, while Q-switch at divided by integer number repetition rate. 0 – repetition rate is divide by pump; 1 – repetition rate is divided by Q-switch.



Table 7: List of the H-SMART harmonic generator control commands

Description	Command
Set SH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "SH-temp": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius.
Set TH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "TH-temp": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius.
Set FH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "FH-temp": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius.
Set FiH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "FiH-temp": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius.
Set default SH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "SH-temp-default": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius. The received value is saved into non-volatile memory.
Set default TH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "TH-temp-default": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius. The received value is saved into non-volatile memory.
Set default FH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "FH-temp-default": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius. The received value is saved into non-volatile memory.
Set default FiH crystal temperature	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "harmonics", "forward-protocol": "crystals", "forward-action": "control", "FiH-temp-default": 45.0 } } }</pre> Temperature should be positive number in xx.yy format. Acceptable range is 30 – 65 deg Celsius. The received value is saved into non-volatile memory.
Autotune harmonic	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-</pre>



Description	Command
crystal	<pre>device":"harmonics","forward-protocol":"crystals","forward-action":"control","autotune-mode":x}}}</pre> <i>"autotune-mode" key value X determines:</i> 2 – initial search for second harmonic optimal temperatures of all crystals. 3 – initial search for third harmonic optimal temperatures of all crystals. 4 – initial search for fourth harmonic optimal temperatures of all crystals. 5 – initial search for fifth harmonic optimal temperatures of all crystals. 6 – fine tuning of second harmonic optimal temperature only. 7 – fine tuning of third harmonic optimal temperature only. 8 – fine tuning of fourth harmonic optimal temperature only. 9 – fine tuning of fifth harmonic optimal temperature only.
Change harmonic unit status	<pre>{"action":"control","code":{"device":"laser","values":{"forward-device":"harmonics","forward-protocol":"crystals","forward-action":"control","hu-status":X}}}</pre> <i>Change harmonic crystals temperatures according preset table.</i> <i>X should be integer number in 1-6 range.</i> 1 – max fundamental output. 2 – max second harmonic output. 3 – max third harmonic output. 4 – max fourth harmonic output. 5 – max fifth harmonic output. 3,6 – configure harmonic unit for OPO pumping with third harmonic. The received value is saved into non-volatile memory.

Table 8: List of the OPO control commands

Description	Command
Set OPO wavelength	<pre>{"action":"control","code":{"device":"laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","wavelength":xxx.yy}}}</pre> <i>Wavelength should be number in 210-2300 range in xxxx.yy format, two decimal points precision is accepted.</i>
Set OPO wavelength in wavenumbers	<pre>{"action":"control","code":{"device":"laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","wavenumber":xxxx}}}</pre> <i>Wavelength can be also set in wavenumber format. Wavenumber should be an integer number in 4350 – 47620 range.</i>
Set attenuation of FF bypass	<pre>{"action":"control","code":{"device":"laser","values":{"forward-device":"harmonics","forward-protocol":"transmission","forward-action":"control","device":"attenuator","transmission":xxx.y}}}</pre> <i>Set attenuation of fundamental frequency (FF) bypass attenuator. 0 value corresponds to full pulse energy going for OPO pumping, 100 – full FF pulse energy going to Port #2, and zero for OPO pumping.</i> <i>Attenuation should be number in 0- 100 range in xxx.yy format, one decimal point precision is accepted.</i>
Set transmission of OPO	<pre>{"action":"control","code":{"device":"laser","values":{"forward-</pre>



Description	Command
attenuator (when optional INTERNAL attenuator is present)	<pre>device:"opo","forward-protocol":"wavelength","forward-action":"control","opo-transmission":xxx.yy}}</pre> Transmission should be number in 0- 100 range in xxx.yy format, one decimal point precision is accepted.
Set transmission of OPO attenuator (when optional EXTERNAL attachable attenuator is present)	<pre>'{"action":"control","code":{"device": "attenuator","values":{"device":"attenuator","transmission":xxx.yy}}}</pre> Transmission should be number in 0- 100 range in xxx.yy format, one decimal point precision is accepted.
Configure laser for OPO pumping	<pre>{"action":"control","code":{"device":"laser","values":{"forward-device":"harmonics","forward-protocol":"crystals","forward-action":"control","hu-status": 3.6}}}</pre> Execute this command for re-configuration of laser for OPO pumping.
Finetune wavelength	<pre>{"action":"control","code":{"device": "laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","finetune-wavelength": X}}}</pre> Changes angle of OPO crystal by X steps, resulting in shift of OPO produced wavelength. Accepted X values are from -5 to 5.
Finetune pulse energy	<pre>{"action":"control","code":{"device": "laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","finetune-energy": X}}}</pre> When two crystal OPO is used, this command changes relative angular position of one OPO crystal, which in turn changes produced pulse energy. Accepted X values are from -5 to 5. NOTE: in 210-410 nm range this command changes angle of second harmonic crystal.
Finetune pointing	<pre>{"action":"control","code":{"device": "laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","finetune-pointing": X}}}</pre> Changes angle of mirror for change of beam pointing in vertical plane. Accepted X values are from -30 to 30.
Finetune phase	<pre>{"action":"control","code":{"device": "laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","finetune-phase": X}}}</pre> Changes angle of phase plate or crystal. Accepted X values are from -5 to 5.
Save wavelength correction to non-volatile memory	<pre>{"action":"control","code":{"device": "laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","correct-wavelength":XX}}}</pre> Save angle deviation from default OPO wavelength tuning lookup table to non-volatile memory. XX values are in microsteps, should be integer number. XX value typically does not exceed ± 50 steps.
Save pulse energy correction to non-volatile memory	<pre>{"action":"control","code":{"device": "laser","values":{"forward-device":"opo","forward-protocol":"wavelength","forward-action":"control","correct-energy":XX}}}</pre> Save angle deviation from maximum pulse energy (power) from default OPO



Description	Command
	<i>tuning lookup table to non-volatile memory. XX values are in microsteps, should be integer number. XX value typically does not exceed ± 50 steps.</i> <i>NOTE: in 210-410 nm range deviation angle of second harmonic crystal is saved.</i>
Save pointing correction to non-volatile memory	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "opo", "forward-protocol": "wavelength", "forward-action": "control", "correct-pointing": XX } } }</pre> <i>Save angle deviation of beam pointing from default lookup table to non-volatile memory. XX values are in microsteps, should be integer number. XX value typically does not exceed ± 50 steps.</i>
Save phase correction to non-volatile memory	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "opo", "forward-protocol": "wavelength", "forward-action": "control", "correct-phase": XX } } }</pre> <i>Save angle deviation of phase plate or crystal from default lookup table to non-volatile memory. XX values are in microsteps, should be integer number. XX value typically does not exceed ± 50 steps.</i>
Save OPO offset to non-volatile memory	<pre>{ "action": "control", "code": { "device": "laser", "values": { "forward-device": "opo", "forward-protocol": "wavelength", "forward-action": "control", "offset-save": XX } } }</pre> <i>Save offset of OPO wavelength tuning lookup table to non-volatile memory. This command should be executed at default wavelength. The lookup table is shifted by XX microsteps, resulting in shift at all wavelengths.</i> <i>XX values are in microsteps, should be integer number. XX value typically does not exceed ± 50 steps.</i>
Configure scan settings	<pre>{ "action": "control", "code": { "device": "laser", "values": { "scan-start-at": 420, "scan-finish-at": 450, "scan-step": 2.0, "scan-pulses-set": 100, "scan-mode-set": 1 } } }</pre> <i>Configures Q-TUNE scan parameters.</i> <i>“scan-start-at”, “scan-finish-at” should be number in 210-2300 nm range in xxx.yy format, two decimal points precision is accepted.</i> <i>“scan-step” should be number in 0.01 – 10 nm range in xx.yy format, two decimal points precision is accepted.</i> <i>“scan-pulses-set” is determining how many pulses are produced at each scan wavelength. The number can be in 1-100000 range. Note, during wavelength change laser operation is disabled and pulses are not counted, also no laser output is produced.</i> <i>“scan-mode-set” is determining scan method. At the moment only “1” is valid number that determines step-by-step scanning mode.</i>
Start wavelength scan	<pre>{ "action": "control", "code": { "device": "laser", "values": { "scan-run": 1 } } }</pre> <i>Starts the scanning of the wavelength. “scan-run” parameter remains 1 until scan is finished, then it is reset to 0. The remaining pulses to be made can be checked in values page, check “scan-pulses-remaining” parameter.</i>



12.3 Decoding Error Codes

Each error, detected by laser control firmware, sets a corresponding flag in error register. The error register is 32-bit long. Value of register in its decimal form can be read from web-interface or from /values? server page, look for a key “error-code”.

Error code can be decoded in the following way. Use decimal to binary decoder like this one: <https://www.rapidtables.com/convert/number/decimal-to-binary.html>.

For example, once you enter 262144 into decoder, you will see that bit #18 is "1" while all remaining bits are "0". Using the table below check that bit #18 corresponds to "Pulse counter error".

Table 9: Description of error flags

Bit	Description
E0	Max pump current exceeded
E1	Laser ID does not match
E2	Pump current configuration error
E3	Could not detect laser ID
E4	DAC write error
E5	ADC read error
E6	Maximum operating temperature exceeded
E7	Temperature sensor malfunction
E8	Humidity sensor malfunction
E9	Pressure sensor malfunction
E10	Power supply undervoltage
E11	Overheat of TEC controller
E12	+12VDC power supply malfunction
E13	Pump diode driver #1 overheat
E14	Pump diode driver #2 overheat
E15	Pump diode driver #1 undervoltage
E16	Pump diode driver #2 undervoltage
E17	General pump diode driver malfunction
E18	Pulse counter error
E19	Laser DISABLE interlock activated
E20	Stepper motor driver error
E21	Burst timeout counter error
E22	Humidity is out of range



E23	External pulse generator error
E24	LD driver temperature is out of range
E25	Pump diode temperature is out of range
E26	Pressure is out of range
E27	Wakeup procedure error
E28	Internal communication error
E29	Laser emergency error
E30	External sensor or/and driver error
E31	External auxiliary equipment error



13 Troubleshooting

Problem:	Solution:
Laser doesn't turn on	Check cables: Check all connections between laser head, controller and power adapter (detailed information can be found in laser manual Chapter 8).
Can't access laser control panel through my personal computer	• Make sure the laser is switched on and STATUS LED is turned yellow (after the warm up). • Check if Ethernet cable connected correctly. Use QLI supplied network cable and USB to Ethernet adapter. If these are not working try to use build in into computer Ethernet socket, or try to switch to another computer. • Check if your computer is configured correctly for the communication with the laser (instructions how to do that can be found in laser manual Chapter 10). • Try using Firefox or Chrome web-browser (Safari web-browser had issues reported in the past). • Make sure, you enter IP address correctly and with port part (e.g: 192.168.1.34:7557/).

If above located solutions do not solve the problem user should contact manufacturer for the instructions. To make this process easier and to help the manufacturer to better understand the encountered problem user should download laser “log file” from the laser control panel. To do this go to laser control panel, select the “Help” window and press “View log file” button (Figure 32). Copy/paste the opened log file and send it to the manufacturer.

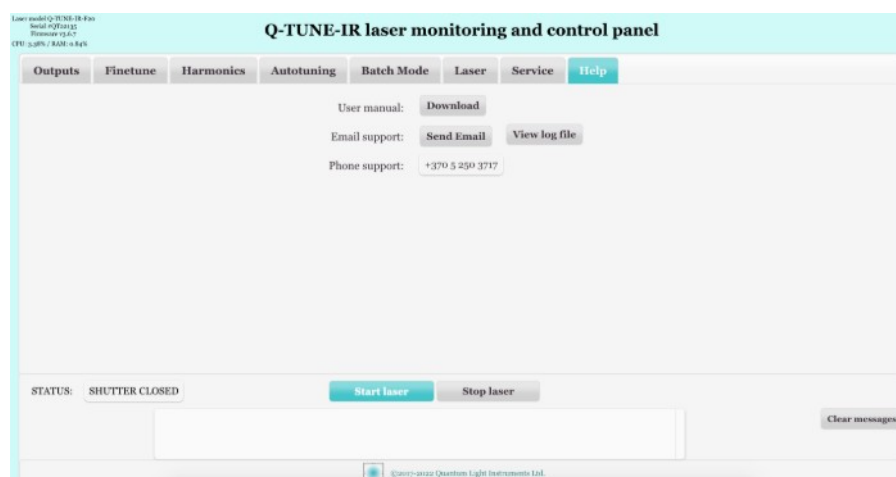


Figure 32: Laser control panel “Help” window



14 Maintenance

14.1 Air Purging Unit (APU)

If requested by the customer our laser systems can be equipped with Air Purging Unit (APU). Purge gasses help to increase the longevity of the laser system by protecting its optical components from two different sources of damage: harmful particles (dust, vapor and other particles that might coat optical components) and condensation.

Air Purging Unit has integrated module with **gas dryer/purifier** units which might require regeneration or replacement. During normal operation APU cartridge can last 200 to 500 hours.

Gas dryer/purifier is filled with indicating material and 5A Molecular Sieves for removal of moisture, impurities and various other harmful materials. Desiccant granules which will dry the air to the dew point of -100°F (-73.3°C) at the atmospheric pressure up to 90 psi. Gas dryer requires **constant maintenance and should be regenerated when desiccant granules become exhausted (Indicator label turns from blue to pale lavender)**. Status of the gas dryer can be checked by looking through holes in the cover of the Air Purging Unit (Figure 33). To increase visibility use the flash light.

Reset “APU run time” counter in the service tab once new/regenerated desiccant cartridge is installed.

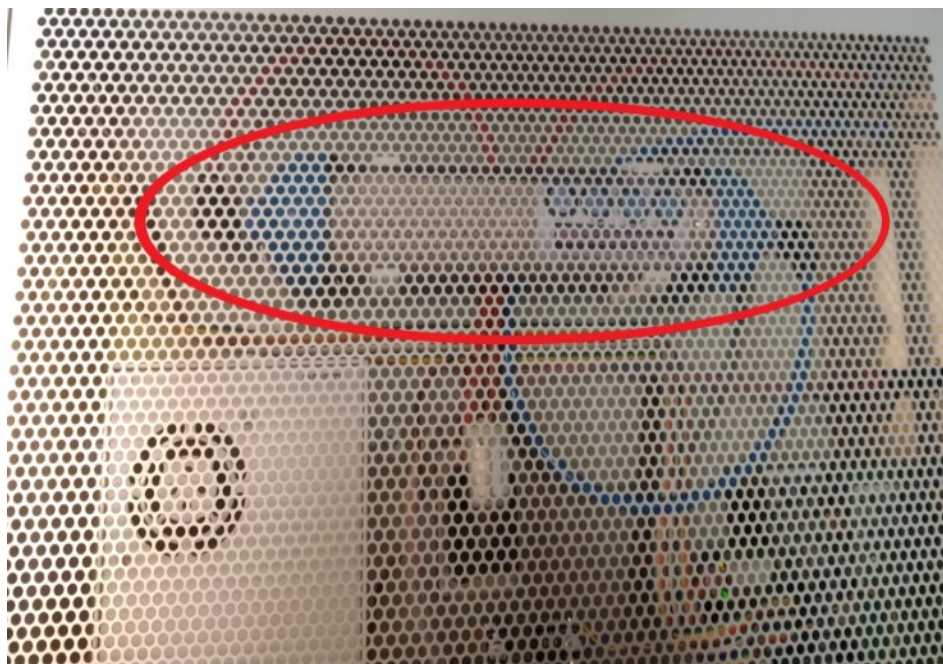


Figure 33: Gas dryer/purifier cassette is visible through APU's perforated cover.

New cartridge can be bought from QLI or representative, cartridge regeneration is also possible:





Figure 34: Gas dryer/purifier with humidity indicator

1. Remove Air dryer/purifier cassette from APU unit, cassette is held in place by a pair of clips.

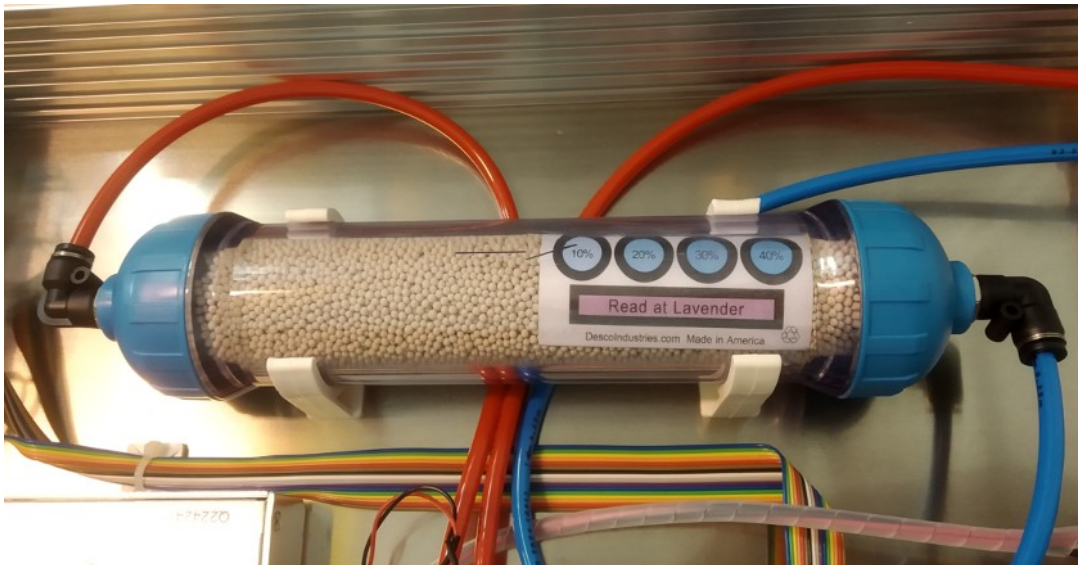


Figure 35: Inside of the Air Purging Unit.

2. Using Torx T3 screwdriver loosen four screws holding top cover (Figure 36). Remove top cover by sliding it outwards..



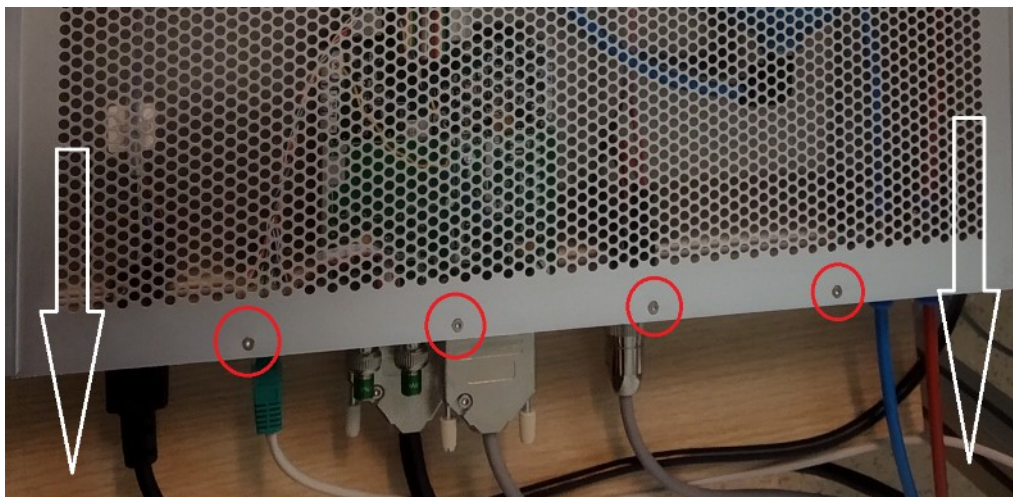


Figure 36: Removing APU unit cover. Retaining screws are marked in red. Cover removal direction is marked by white arrows.

3. Remove cassette tube from the holder and replace with the new one or regenerate the granules:
4. Unscrew one of two blue cartridge caps. Spread granules in layers one granule deep and heat for 1 hour at 425°F (210°C). **Regenerated granules should be cooled down in closed glass or metal container down before being put back in their tube (hot regenerated materials might damage the plastic tube).** Gas dryer granules indicating color might become less distinct after successive regenerations. The granules should not be overheated. High temperatures might alter the materials crystal structure what in turn will turn desiccant granules permanently inactive. **Do not heat the indicating label.**
5. Put regenerated desiccant granules back in the tube.
6. Repeat steps 1-4 in reverse order to assemble module.

